

# Completeness Extraction for $\mathcal{ALCI}_{\text{RCC5}}$ (v2.1, GPT-5.5 v2-Review Repairs)

Closing the model-to-quotient gap via witness-generated submodels,  
generated covers, reachability discharge, and explicit equality data

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**Changelog v2 → v2.1 (May 2026).** GPT-5.5 reviewed the v2 manuscript and identified six remaining proof obligations [8]. This revision discharges *all six*:

- **(O1) Blocking vs. semantic EQ separation.** The typed-EQ colour is replaced by an *explicit finite equality relation*  $\equiv_{\text{EQ},Q}$  on *equality ports* of the certificate. Profile equality alone never implies  $\equiv_{\text{EQ}}$ . Mosaic axiom (M7) is rewritten: *EQ-label iff endpoints are linked by  $\equiv_{\text{EQ},Q}$  in that mosaic*. An explicit *vertical separation* axiom forbids  $\equiv_{\text{EQ},Q}$  along strict PP/PPI-paths. See Section 10.
- **(O2) EQ-aware reachability discharge.** Reachability quantification is lifted from a single occurrence to the EQ-mate class. Existential:  $\exists R.D \in \lambda_Q(\sigma) \Rightarrow \exists \hat{\sigma} \in \text{Mate}(\sigma) \exists \sigma' (\hat{\sigma} \rightsquigarrow_R \sigma' \wedge D \in \lambda_Q(\sigma'))$ . Universal:  $\forall R.D \in \lambda_Q(\sigma) \Rightarrow \forall \hat{\sigma} \in \text{Mate}(\sigma) \forall \sigma' (\hat{\sigma} \rightsquigarrow_R \sigma' \Rightarrow D \in \lambda_Q(\sigma'))$ . This handles the multiple-incomparable-upward-witness stress formula  $C_{AB}$ . See Section 13.
- **(O3) EQ-modality normalization.** A preliminary preprocessing lemma eliminates  $\exists \text{EQ}.D$  and  $\forall \text{EQ}.D$  by  $\exists \text{EQ}.D \equiv D$  and  $\forall \text{EQ}.D \equiv D$  (sound under strong EQ semantics). After preprocessing, all closure-existentials and closure-universals range over  $\{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\} = \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$  only. See Section 2.
- **(O5) DR/PO side-context witness lemma.** A formal lemma (Lemma 9.5) ensures that every DR/PO witness placed by the witness policy lives in a bounded side context  $\text{Side}(u, v)$  of size  $\leq 5$ , that the realized mosaic on this side context satisfies all (M1)–(M7), and that all relation-safety, universal-firing, and RCC5 composition constraints at the interface are preserved. See Section 9.1.
- **(O6) Simultaneous eventuality realization.** A canonical-unfolding theorem (Lemma 14.6) shows that the split tree  $T$  from Construction 6.1 is itself a concrete realization of the canonical unfolding  $U(Q)$  in which *all* PP/PPI-eventualities at *all* occurrences are realized simultaneously: (SR1) PP via the ultimately periodic parent walk, (SR2) PPI via child multiplicity, (SR3) universal propagation via Lemma 13.5. Mate-class quantification (introduced for O2) is essential for simultaneity. Patchwork [1] ensures that the simultaneously realized witnesses admit a coherent atomic RCC5 refinement. See Section 14.
- **(O4) Finite mosaic arity bound.** A locality theorem (Theorem 12.4) establishes the effective constant  $N(C_0) = 5$ : every mosaic axiom (M1)–(M7) is  $\leq 3$ -local (Lemma 12.1), every support-closure operation (SC1)–(SC4) is  $\leq 5$ -local (Lemma 12.2), and by the patchwork property of RCC5 [1] a labelled complete graph of any arity is in a support-closed descriptor iff every sub-network of arity  $\leq 5$  is. Certificate enumeration is bounded by  $|E_d|^5 \cdot |\{\text{DR}, \text{PO}, \text{EQ}, \text{PP}, \text{PPI}\}|^{(5)}$ , an effectively computable function of  $|\text{cl}(C_0)|$  (Corollary 12.5). See Section 12.

With all six obligations discharged, the revised proof closes the completeness extraction without remaining acknowledged gaps. See Section 17 for honest scope.

## Abstract

This is a repaired version of the completeness-extraction note for the split-forest approach

to  $\mathcal{ALCI}_{\text{RCC5}}$  decidability. An earlier version [9] contained two structural defects that prevented it from establishing the model-to-quotient direction: a Hasse construction that assumed a well-founded ambient  $\leq_{\text{PP}}$  DAG with a  $\leq_{\text{PP}}$ -minimal root (which fails for concepts forcing infinite PP-chains, e.g.  $\exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$ ), and a triple-coherence proof with witness conflation across distinct occurrences.

The repair follows the structural ideas of GPT 5.5’s coherent generated split-forest proof [6], but *without* introducing Büchi automata or  $\omega$ -acceptance machinery. We apply seven concrete repairs:

- (R1) Pass to a *witness-generated submodel* (sparse, logic-generated, no density issue);
- (R2) Use *generated cover edges* (selected witness-cover edges in the generated submodel), not semantic Hasse covers;
- (R3) Drop the ambient-root requirement; permit *self-loops on the parent function* of the finite quotient, so ultimately periodic ancestor trunks are first-class;
- (R4) Make the *containment orientation* explicit (parent = proper superpart, child = proper part);
- (R5) Replace raw pair-tables with *support-closed mosaics* that store joint atomic networks, fixing the C5 witness conflation;
- (R6) Add an explicit *typed-EQ congruence* axiom, separate from blocking equivalence;
- (R7) Discharge transitive PP/PPI eventualities by *finite-graph reachability* on the finite quotient.

The crucial observation is that the cycle-realization condition of [6] reduces to ordinary reachability over the finite quotient:  $\exists \text{PP}.D \in \lambda(\sigma)$  requires  $D \in \lambda(\sigma')$  for some  $\sigma'$  reachable from  $\sigma$  along parent edges (including self-loops). This is decidable in  $O(|S|^2)$  time per eventuality and requires no  $\omega$ -language machinery.

Combined with GPT-5.4’s soundness chain [5], this yields decidability of concept satisfiability in  $\mathcal{ALCI}_{\text{RCC5}}$  under strong EQ semantics.

## 1 Introduction

The split-forest proposal for  $\mathcal{ALCI}_{\text{RCC5}}$  decidability [5] proves the soundness direction in full detail: every valid finite rank- $d$  quotient unfolds into a genuine strong-EQ model via canonical refinements of disjunctive RCC5 networks. The completeness direction—that every satisfiable concept admits a valid finite quotient—was stated as a proof sketch.

A first attempt to write out this completeness extraction (“v1”) [9] appeared in April 2026. Subsequent review by GPT-5.5 identified two structural defects:

**Defect 1: Hasse construction with rooted-tree assumption.** v1 began by orienting the proper-part relation  $\leq_{\text{PP}}$  as a DAG and choosing a  $\leq_{\text{PP}}$ -minimal element as ambient root. This works when the model has a unique minimum or when one can be added freshly. But the concept

$$C \uparrow := \exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$$

forces a strictly increasing PP-chain  $d_0 \text{PP} d_1 \text{PP} d_2 \text{PP} \dots$  with no  $\leq_{\text{PP}}$ -minimum. Adding a fresh top via  $\rho(d_\top, d) = \text{PP}$  *inverts* the orientation: by RCC5 semantics,  $d_\top \text{PP} d$  means  $d_\top$  is a proper

part of  $d$ , not the other way around.  $v_1$ 's construction therefore breaks on any concept forcing an infinite PP-chain.

**Defect 2: Witness conflation in (C5) triple coherence.**  $v_1$ 's proof of (C5) implicitly combined witnesses from different occurrences as if they formed a coherent triple. Raw pair tables admit independent witnesses for  $M(\chi_1, \chi_2)$  and  $M(\chi_1, \chi_3)$  that may not be jointly realizable. Triple coherence requires a single triple  $(x_1, x_2, x_3)$  realizing all three relations simultaneously; raw tables do not enforce this.

GPT-5.5's coherent generated split-forest proof [6] repairs both defects. We follow its approach with one important difference: where GPT-5.5 phrases the eventuality discharge in automata-theoretic language ("cycle-realization condition", Definition 6.4 of that paper), we observe that this condition reduces to *finite-graph reachability over the finite quotient*, and we present the proof in that finitary style.

## 1.1 Plan

Section 2 eliminates EQ-modalities by preprocessing, so the rest of the proof handles only  $\{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\} = \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$  existentials and universals. Section 3 recaps types, closure, need families, and the model-edge relation-safety lemma. Section 4 introduces the witness-generated submodel reduction. Section 5 fixes the containment orientation and defines generated cover edges. Section 6 constructs the generated split-tree presentation, with no ambient root and self-loops permitted on the parent function. Section 7 defines rank- $d$  profiles and the finite quotient. Section 8 extracts support-closed sibling-interface mosaics. Section 9 defines witness menus for DR/PO existentials. Section 10 states the typed-EQ congruence axiom. Section 11 verifies (C1)–(C5). Section 13 proves reachability discharge for PP/PPI eventualities. Section 15 assembles validity. Section 16 states the main theorem and decidability. Section 17 compares with GPT-5.5's automata version and discusses honest scope.

## 1.2 Main result

**Theorem 1.1** (Completeness extraction). *If  $\mathcal{I}, d_* \models C_0$  in  $\mathcal{ALCI}_{\text{RCC5}}$  under strong EQ semantics, then there exists a valid finite generated split-forest quotient  $Q$  for  $C_0$  with  $d = \text{md}(C_0)$ .*

Combined with the soundness chain of [5, 6], this yields:

**Corollary 1.2** (Decidability of  $\mathcal{ALCI}_{\text{RCC5}}$ ). *Concept satisfiability in  $\mathcal{ALCI}_{\text{RCC5}}$  under strong EQ semantics is decidable.*

## 2 EQ-modality normalization (preprocessing)

Under strong EQ semantics, EQ is interpreted as the identity relation:  $\rho(a, b) = \text{EQ}$  iff  $a = b$ . Hence EQ as a modal role is degenerate. We absorb it before the rest of the extraction runs.

**Lemma 2.1** (EQ-modality elimination, strong EQ). *Under strong EQ semantics, for every  $\mathcal{ALCI}_{\text{RCC5}}$  concept  $D$ :*

$$\exists \text{EQ}.D \equiv D, \quad \forall \text{EQ}.D \equiv D.$$

*Proof.* Strong EQ semantics fixes  $\text{EQ}^{\mathcal{I}} = \{(a, a) : a \in \Delta^{\mathcal{I}}\}$ .  $\mathcal{I}, a \models \exists \text{EQ}.D$  iff some  $b$  with  $\rho(a, b) = \text{EQ}$  has  $b \in D^{\mathcal{I}}$ , iff  $a \in D^{\mathcal{I}}$ . Dually,  $\mathcal{I}, a \models \forall \text{EQ}.D$  iff every  $b$  with  $\rho(a, b) = \text{EQ}$  satisfies  $D$ , iff  $a \in D^{\mathcal{I}}$  (the only such  $b$  is  $a$  itself).  $\square$

**Construction 2.2** (EQ-normal form). Given an  $\mathcal{ALCI}_{\text{RCC5}}$  concept  $C$  in NNF, define  $C^{-\text{EQ}}$  by recursive replacement: every subterm of the form  $\exists \text{EQ}.D$  or  $\forall \text{EQ}.D$  (after recursively normalizing  $D$ ) is replaced by  $D^{-\text{EQ}}$ . All Boolean and non-EQ-modality constructors are preserved.

**Proposition 2.3** (Soundness of normalization).  $C \equiv C^{-\text{EQ}}$  under strong EQ semantics, and the size of  $C^{-\text{EQ}}$  is at most  $|C|$  (each replacement strictly removes a modal layer).

*Proof.* Lemma 2.1 applied at each replacement, by induction on subterm depth, with NNF preservation by the symmetry  $\exists \text{EQ} \leftrightarrow \forall \text{EQ}$ .  $\square$

*Remark 2.4* (Standing assumption from now on). For the remainder of this paper we assume that  $C_0 = C_0^{-\text{EQ}}$ , i.e. no EQ-modality appears anywhere in  $\text{cl}(C_0)$ . Hence every existential and universal in the closure has role  $R \in \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\} = \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$ . This eliminates a third case in every per-relation argument that follows; in particular the validity proof (V1, V2) needs no  $R = \text{EQ}$  branch.

(*Aside.* Under weak EQ semantics, the above lemma fails: EQ may have non-trivial mate classes. The standard TBox-internalization reduction from weak to strong EQ still applies before normalization, so the strong-EQ assumption costs nothing in expressivity.)

### 3 Preliminaries

We work under the conventions of [5]. This section recalls only the notions used in the extraction.

#### 3.1 Types and closure

Fix a concept  $C_0$  in negation normal form (NNF). The Fischer-Ladner closure  $\text{cl}(C_0)$  is the smallest set containing  $C_0$  closed under subformulas and NNF complements. A *Hintikka type*  $\tau \in \text{Tp}(C_0) \subseteq 2^{\text{cl}(C_0)}$  is a maximally consistent subset satisfying the standard propositional closure conditions. For  $a$  in a model  $\mathcal{I}$ , write  $\text{tp}_{\mathcal{I}}(a) = \{D \in \text{cl}(C_0) : \mathcal{I}, a \models D\}$ ; this is a Hintikka type.

#### 3.2 Need families and relation-safety

**Definition 3.1** (Per-relation needs). For  $\tau \in \text{Tp}(C_0)$  and  $R \in \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$ , define

$$\text{Need}_R(\tau) := \{D \in \text{cl}(C_0) : \forall R.D \in \tau\}.$$

A typed edge  $(\tau, R, \sigma)$  with  $\tau, \sigma \in \text{Tp}(C_0)$  is *relation-safe* when

$$\text{Need}_R(\tau) \subseteq \sigma \quad \text{and} \quad \text{Need}_{\text{inv}(R)}(\sigma) \subseteq \tau.$$

**Lemma 3.2** (Model edges are relation-safe). In any model  $\mathcal{I}$  of  $\mathcal{ALCI}_{\text{RCC5}}$ , if  $\rho(a, b) = R \in \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$  for distinct elements  $a, b$ , then  $(\text{tp}_{\mathcal{I}}(a), R, \text{tp}_{\mathcal{I}}(b))$  is relation-safe.

*Proof.* If  $\forall R.D \in \text{tp}_{\mathcal{I}}(a)$ , then  $\mathcal{I}, a \models \forall R.D$  and  $\rho(a, b) = R$ , hence  $\mathcal{I}, b \models D$ , so  $D \in \text{tp}_{\mathcal{I}}(b)$ . The dual inclusion follows by applying the same argument with the inverse relation at  $b$ .  $\square$

This elementary observation is the engine of the entire extraction: every relation realized in the model is automatically relation-safe for its endpoints' types.

### 3.3 RCC5 composition

We use the standard RCC5 composition table  $\text{comp}(R, S) \subseteq \{\text{DR}, \text{PO}, \text{EQ}, \text{PP}, \text{PPI}\}$  (see [6, Table 1]). The composition is sound for any abstract RCC5 frame:  $\rho(a, c) \in \text{comp}(\rho(a, b), \rho(b, c))$ .

## 4 Witness-generated submodel reduction

The first repair is to restrict attention to sparse, logic-generated submodels. This sidesteps every concern about arbitrary-density semantic orders and lets us define cover edges relative to a finite generated set.

**Definition 4.1** (Selected witness policy). Let  $\mathcal{I}$  be a model. A *selected witness policy*  $w$  chooses, for every  $a \in \Delta^{\mathcal{I}}$  and every existential  $\exists R.D_0 \in \text{cl}(C_0)$  with  $a \in (\exists R.D_0)^{\mathcal{I}}$ , one element

$$w(a, \exists R.D_0) \in \Delta^{\mathcal{I}} \quad \text{with} \quad \rho(a, w(a, \exists R.D_0)) = R \text{ and } w(a, \exists R.D_0) \in D_0^{\mathcal{I}}.$$

We write  $a \xrightarrow{R}_w b$  when  $b$  is so selected.

**Definition 4.2** (Witness-generated submodel). Let  $a_0 \in \Delta^{\mathcal{I}}$ . The *witness-generated domain*  $D_w(a_0) \subseteq \Delta^{\mathcal{I}}$  is the smallest set containing  $a_0$  and closed under all selected witnesses for closure-existentials true at its members. The *induced submodel*  $\mathcal{I}_w$  has domain  $D_w(a_0)$  with  $\rho^{\mathcal{I}_w}$  and atomic concept interpretations restricted from  $\mathcal{I}$ .

**Lemma 4.3** (Witness-generated submodel lemma). *If  $a_0 \in C_0^{\mathcal{I}}$ , then for every  $a \in D_w(a_0)$  and every  $E \in \text{cl}(C_0)$ ,*

$$a \in E^{\mathcal{I}} \iff a \in E^{\mathcal{I}_w}.$$

*In particular,  $a_0 \in C_0^{\mathcal{I}_w}$ .*

*Proof.* The induced substructure remains a strong abstract RCC5 frame: all RCC5 frame requirements (irreflexivity of  $\{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$ , converse, composition) are universal over retained pairs and triples and are preserved under restriction. Atomic concepts are preserved by restriction.

Boolean cases of the equivalence are immediate from the inductive hypothesis.

For  $E = \exists R.F$ : if  $a \in E^{\mathcal{I}_w}$ , retention into  $\mathcal{I}_w$  implies the witness exists in  $\mathcal{I}$ , so  $a \in E^{\mathcal{I}}$ . Conversely, if  $a \in E^{\mathcal{I}}$ , the selected witness  $b = w(a, \exists R.F)$  lies in  $D_w(a_0)$  and satisfies  $F$  in  $\mathcal{I}$ ; by induction  $b \in F^{\mathcal{I}_w}$ , so  $a \in E^{\mathcal{I}_w}$ .

For  $E = \forall R.F$ : if  $a \in E^{\mathcal{I}}$ , every retained  $R$ -neighbour of  $a$  is also an  $R$ -neighbour in  $\mathcal{I}$  and satisfies  $F$ ; by induction it satisfies  $F$  in  $\mathcal{I}_w$ . Conversely, suppose  $a \notin E^{\mathcal{I}}$ . Then  $a \in (\exists R.\bar{F})^{\mathcal{I}}$ , where  $\bar{F}$  is the NNF complement, and  $\exists R.\bar{F} \in \text{cl}(C_0)$  because the closure is closed under NNF complements. The selected witness for  $\exists R.\bar{F}$  is retained and, by induction, satisfies  $\bar{F}$  in  $\mathcal{I}_w$ ; hence  $a \notin E^{\mathcal{I}_w}$ .  $\square$

**Corollary 4.4** (Sparse generated models suffice). *Every satisfiable  $C_0$  has a witness-generated model with  $|D_w(a_0)| \leq \aleph_0$ . The cover-tree extraction below need only address this sparse class.*

*Remark 4.5* (Why the density issue dissolves). Lemma 4.3 replaces v1’s flawed Hasse construction with a model-theoretic thinning. The resulting  $\mathcal{I}_w$  contains exactly the elements that the logic witnesses for: the evaluation point and selected witnesses for existentials in the closure. Density of an arbitrary semantic PP/PPI order is irrelevant, because we never quantify over it.

## 5 Generated cover edges and containment orientation

We now fix the orientation convention used throughout the rest of the paper.

**Containment orientation.** *Parent* = proper superpart; *child* = proper part.  
Equivalently: along a tree branch, child PP parent and parent PPI child.

This convention determines which existentials produce “upward” moves and which produce “downward” moves in the split tree.  $\exists\text{PP}.D$  at  $a$  requires a proper-superpart witness, hence an *ancestor* in the tree.  $\exists\text{PPI}.D$  at  $a$  requires a proper-part witness, hence a *descendant*.

**Definition 5.1** (Generated cover edge). Let  $A \subseteq D_w(a_0)$  be a finite witness-generated subset of the submodel. A *generated PPI-cover edge*  $x \xrightarrow{\text{PPI}}_{\text{gen},A} y$  holds when:

- $x, y \in A$  and  $\rho(x, y) = \text{PPI}$  (i.e.,  $y$  is a proper part of  $x$ );
- no element  $z \in A$  satisfies  $\rho(x, z) = \text{PPI}$  and  $\rho(z, y) = \text{PPI}$  (no selected element of  $A$  lies strictly between  $x$  and  $y$  in the proper-part order).

The dual generated PP-cover edge  $x \xrightarrow{\text{PP}}_{\text{gen},A} y$  is defined symmetrically.

*Remark 5.2* (Generated covers are not Hasse covers). A generated cover edge is immediate relative to the finite selected set  $A$ . It is not claimed to be an immediate cover in the full ambient PP-order. Repaired wording from the survival note [7]:

*A cover edge is a generated witness-cover edge, not a semantic Hasse cover.*

Constructing covers relative to  $A$  is well-defined even when the ambient order is dense or has no Hasse covers at all; we simply pick the immediate-in- $A$  relation. Crucially,  $A$  grows with the generation depth, but at every finite depth the relative cover relation is well-defined.

## 6 Generated split-tree presentation

We construct a (possibly infinite) split tree from a witness-generated model. No ambient root is required.

**Construction 6.1** (Generated split-tree). Let  $\mathcal{I}_w$  be a witness-generated model with evaluation point  $a_0$ . Build a tree  $T$  of *occurrences* together with maps

$$\text{orig} : T \rightarrow D_w(a_0), \quad \lambda : T \rightarrow \text{Tp}(C_0), \quad \text{par} : T \rightharpoonup T,$$

inductively as follows.

1. **Seed.** Create a root occurrence  $u_0$  with  $\text{orig}(u_0) = a_0$  and  $\lambda(u_0) = \text{tp}_{\mathcal{I}_w}(a_0)$ .
2. **Upward expansion (PP-witnesses).** For each occurrence  $u$  and each  $\exists \text{PP}.D \in \lambda(u)$ :
  - If  $\text{par}(u)$  is undefined or  $\lambda(\text{par}(u)) \not\supseteq D$ , select a witness  $b = w(\text{orig}(u), \exists \text{PP}.D) \in D_w(a_0)$  with  $\rho(\text{orig}(u), b) = \text{PP}$  and create a new occurrence  $u^\uparrow$  with  $\text{orig}(u^\uparrow) = b$ ,  $\lambda(u^\uparrow) = \text{tp}_{\mathcal{I}_w}(b)$ ,  $\text{par}(u) = u^\uparrow$ .
  - If a parent already records  $D$ , no new occurrence is needed.
3. **Downward expansion (PPI-witnesses).** For each occurrence  $u$  and each  $\exists \text{PPI}.D \in \lambda(u)$ , select a witness  $b = w(\text{orig}(u), \exists \text{PPI}.D)$  and create a child occurrence  $u^\downarrow$  with  $\text{orig}(u^\downarrow) = b$ ,  $\lambda(u^\downarrow) = \text{tp}_{\mathcal{I}_w}(b)$ , and  $\text{par}(u^\downarrow) = u$ .
4. **Sibling expansion (DR/PO-witnesses).** For each occurrence  $u$  and each  $\exists R.D \in \lambda(u)$  with  $R \in \{\text{DR}, \text{PO}\}$ , select a witness  $b = w(\text{orig}(u), \exists R.D)$  and record an outgoing  $R$ -witness edge from  $u$  to a node mapped to  $b$  (typically a sibling of  $u$  under a common ancestor).
5. **Splitting.** If a single element  $b \in D_w(a_0)$  must appear under multiple distinct parents in this construction, introduce one occurrence per containment branch. These occurrences are declared EQ-mates (typed equivalence classes under the relation defined in Section 10).
6. **Edge labels.** For comparable occurrences  $u, v$  on the same branch with  $v = \text{par}^k(u)$  for some  $k > 0$ , set  $\rho_T(u, v) = \text{PP}$  and  $\rho_T(v, u) = \text{PPI}$  (transitively closed along the branch). For incomparable occurrences,  $\rho_T(u, v)$  is determined by the realized RCC5 relation  $\rho(\text{orig}(u), \text{orig}(v))$  in  $\mathcal{I}_w$ .

*Remark 6.2* (Branches need not terminate). The parent chain of an occurrence may be:

- finite, terminating at an occurrence with no  $\exists \text{PP}.D$  in its type;
- one-sided infinite (e.g., for  $C^\uparrow$ );
- bi-infinite (e.g., for concepts forcing both unbounded PP and unbounded PPI chains).

We do *not* require a  $\leq_{\text{PP}}$ -minimal element or any ambient top. In the finite quotient (Section 7), infinite parent chains are represented by self-loops or longer cycles on the parent function.

**Lemma 6.3** (Existence). *Every witness-generated model  $(\mathcal{I}_w, a_0)$  admits a generated split tree as in Construction 6.1.*

*Proof.* Construct occurrences coinductively. At every step, the existing occurrences are finite (since each step adds finitely many witnesses for the closure-existentials of one type). The construction is well-defined because  $\text{cl}(C_0)$  is finite, and there are only finitely many distinct closure-existentials at any type. Pull-back of  $\rho$  via  $\text{orig}$  inherits converse and composition. Type assignments inherit Hintikka conditions from  $\mathcal{I}_w$  via Lemma 4.3.  $\square$

## 7 Rank- $d$ profiles and the finite quotient

We now define the finite combinatorial object that summarizes the (possibly infinite) split tree.



**Definition 7.1** (Profile components). For an occurrence  $u \in T$  at rank  $k$ , the *rank- $k$  profile*  $\text{st}_k(u)$  records:

1. the closure type  $\lambda(u)$ ;
2. the *equality-port tuple*  $\pi(u) = (\pi_1(u), \dots, \pi_n(u))$ , drawn from a finite equality alphabet  $\Pi_{\text{st}_d(u)}$  (Definition 10.1, v2.1). Profile equality on item 2 is the *port alphabet* that an occurrence may carry; semantic EQ-mate equality is recorded separately by the explicit equality relation  $\equiv_{\text{EQ},Q}$  of Definition 10.1 and is *not* implied by profile equality. See Remark 10.10;
3. the rank- $(k-1)$  profile of  $\text{par}(u)$  (with  $\perp$  if undefined);
4. the rank- $(k-1)$  *child multiplicities*: a map  $\Sigma_{k-1} \rightarrow \{0, 1, 2+\}$  counting children by their rank- $(k-1)$  profile;
5. the rank- $(k-1)$  *incident sibling-interface mosaics* for ordered pairs of children (Section 8);
6. the rank- $(k-1)$  *witness menu* for DR/PO existentials (Section 9);
7. the *vertical eventuality summary*: for each  $\exists \text{PP}.D \in \lambda(u)$ , the set of ranks at which  $D$  appears in the parent chain; symmetric for  $\exists \text{PPI}.D$ .

A *rank-0 profile* is just a Hintikka type. The finite profile alphabet  $\Sigma_d$  at rank  $d = \text{md}(C_0)$  has bounded cardinality (Lemma 7.2).

**Lemma 7.2** (Finite-index). *For each  $k \leq d$ , the set  $\Sigma_k$  of rank- $k$  profiles is finite, and  $|\Sigma_k|$  is bounded by an effectively computable function of  $|\text{cl}(C_0)|$ .*

*Proof.* Identical to the finite-index argument in [6, Lemma 7.4]: closure types are finite; equality-port tuples are drawn from a finite alphabet whose size is bounded by an effectively computable function of  $|\text{cl}(C_0)|$  (see Lemma 10.11); child multiplicities are truncated at  $2+$  (threshold counting); incident mosaics are bounded sets of bounded networks over a finite endpoint-summary alphabet (Section 8); witness-menu entries are pairs from a finite set; eventuality summaries are subsets of  $\text{cl}(C_0)$ . Induction on  $k$  closes the bound.  $\square$

**Definition 7.3** (Finite quotient). The *finite generated split-forest quotient* of  $T$  is

$$Q = (S, \lambda_Q, \text{par}_Q, \text{Child}_Q, \text{Iface}_Q, \text{Wit}_Q, \Pi_Q, \equiv_{\text{EQ},Q}, \sigma_*)$$

where:

- $S = \{\text{st}_d(u) : u \in T\} \subseteq \Sigma_d$ ;
- $\lambda_Q(\sigma)$  is the closure type at  $\sigma$ ;
- $\text{par}_Q : S \rightarrow S$  is the projected parent function:  $\text{par}_Q(\sigma) = \text{st}_d(\text{par}(u))$  for any  $u \in T$  with  $\text{st}_d(u) = \sigma$  (well-defined by profile equality, including the parent slot;  $\text{par}_Q(\sigma) = \sigma$  is permitted, as is  $\text{par}_Q(\sigma) = \perp$ );
- $\text{Child}_Q, \text{Iface}_Q, \text{Wit}_Q$  are inherited from the profile components;
- $\Pi_Q = (\Pi_\sigma)_{\sigma \in S}$  is the family of finite equality-port alphabets, and  $\equiv_{\text{EQ},Q}$  is the explicit equality relation on profile-port pairs (Definition 10.1, v2.1);

- $\sigma_* = \text{st}_d(u_*)$  for the occurrence  $u_*$  with  $\text{orig}(u_*) = a_0$ .

*Remark 7.4* (Self-loops are allowed). The parent function  $\text{par}_Q$  may have self-loops:  $\text{par}_Q(\sigma) = \sigma$ . This represents an infinite upward PP-chain that, at rank  $d$ , has stabilized to a single profile. The unfolding interprets each traversal of the self-loop as a fresh occurrence. Concretely, for  $C \uparrow= \exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$ , the quotient has a single profile  $\sigma$  with  $\text{par}_Q(\sigma) = \sigma$  and  $\lambda_Q(\sigma) \ni \exists \text{PP}.\top$ ,  $\lambda_Q(\sigma) \ni \forall \text{PP}.\exists \text{PP}.\top$ .

**Crucially**,  $\text{par}_Q(\sigma) = \sigma$  **does not mean** “ $\sigma$  is its own parent in the model.” It means: the parent occurrence has the same rank- $d$  profile. In the unfolding, every traversal produces a distinct semantic occurrence. This is a blocking cycle, not an EQ-identification.

**Lemma 7.5** (Finiteness of Q).  $|S| \leq |\Sigma_d| < \infty$ .

## 8 Sibling-interface descriptors as support-closed mosaics

This section replaces v1’s raw pair-tables with support-closed mosaics, fixing the C5 witness conflation.

### 8.1 Status partition

For an occurrence  $u$  with two distinct child subtrees  $T_s, T_t$  at children  $s, t$ , we partition descendants of  $s$  by their RCC5 relation to  $t$ :

**Definition 8.1** (Status partition). For  $x$  a descendant of  $s$  (including  $s$  itself):

- $\text{stat}(x) = \text{Core}$  if  $\text{orig}(x)$  lies below both  $\text{orig}(s)$  and  $\text{orig}(t)$  in the generated PPI-order, or is an EQ-mate visible to both sides;
- $\text{stat}(x) = \text{Out}$  if  $x \notin \text{Core}$  and  $\rho(\text{orig}(x), \text{orig}(t)) = \text{DR}$ ;
- $\text{stat}(x) = \text{Front}$  if  $x \notin \text{Core}$  and  $\rho(\text{orig}(x), \text{orig}(t)) = \text{PO}$ .

**Lemma 8.2** (Three-way partition). *Every descendant  $x$  of  $s$  has exactly one status in  $\{\text{Core}, \text{Out}, \text{Front}\}$ . Symmetrically for descendants of  $t$ .*

*Proof.* By Lemma 8.3 and the open cross-edge restriction of [5, Theorem 1.4]: if  $x$  is a non-core descendant of  $s$ , then  $\rho(\text{orig}(x), \text{orig}(t)) \in \{\text{DR}, \text{PO}\}$  (it cannot be PP, PPI, or EQ without forcing  $x$  into **Core** via generated containment closure).  $\square$

**Lemma 8.3** (Open cross-edge). *If  $x$  is a non-core descendant of  $s$  and  $y$  is a non-core descendant of  $t$ , then  $\rho(\text{orig}(x), \text{orig}(y)) \in \{\text{DR}, \text{PO}\}$ .*

*Proof.* EQ would force both into the typed-EQ congruence and hence into **Core**. PP would force  $x$  into the generated PP-closure of material below  $\text{orig}(t)$ , hence visible to  $t$ , contradicting non-coreness. PPI is symmetric.  $\square$

## 8.2 Endpoint summaries and atomic mosaics

**Definition 8.4** (Endpoint summary). An *endpoint summary* at rank  $k$  consists of:

1. a rank- $k$  profile  $p \in \Sigma_k$ ;
2. a status  $\sigma \in \{\text{Core}, \text{Out}, \text{Front}\}$ ;
3. an equality-port tuple  $\pi \in \Pi_p$  (Definition 10.1, v2.1).

The set  $E_k$  of endpoint summaries is finite. Visible equality between two endpoint summaries on the same mosaic is determined by the explicit relation  $\equiv_{\text{EQ},Q}$  (Definition 10.2), not by alphabet coincidence.

**Definition 8.5** (Atomic support mosaic). An *atomic support mosaic* on a finite set  $V$  of vertices, each labelled by an endpoint summary, is a complete labelling  $\mu : V \times V \rightarrow \{\text{DR}, \text{PO}, \text{EQ}, \text{PP}, \text{PPI}\}$  such that:

- (M1)  $\mu(v, v) = \text{EQ}$  for all  $v$ ;
- (M2)  $\mu(w, v) = \text{inv}(\mu(v, w))$ ;
- (M3)  $\mu(u, w) \in \text{comp}(\mu(u, v), \mu(v, w))$  (RCC5 composition);
- (M4) every edge is relation-safe for the closure types of its endpoint summaries (Definition 3.1);
- (M5) Out-status endpoints have only DR as cross-status outgoing label;
- (M6) non-core Front–Front cross-status edges have labels in  $\{\text{DR}, \text{PO}\}$ ;
- (M7) (*M7, v2.1.*) for all  $v, w \in V$ ,  $\mu(v, w) = \text{EQ}$  iff  $v \equiv_{\text{EQ},Q}^\mu w$  (visible equality on  $\mu$  in the sense of Definition 10.2). Profile coincidence alone is *not* sufficient; equality must be witnessed by an explicit  $\equiv_{\text{EQ},Q}$  entry.

**Definition 8.6** (Sibling descriptor). A *rank- $k$  sibling descriptor* for an ordered child pair  $(s, t)$  is a finite set  $\text{Mos}_k(s, t)$  of atomic support mosaics over endpoint summaries at rank  $k$ , capturing the joint atomic networks that occur in the model among finitely many endpoints in  $T_s \cup T_t \cup \{s, t\}$ .

The descriptor is *support-closed* when it is closed under:

- (SC1) **Restriction.** Every induced submosaic on a vertex subset is in  $\text{Mos}_k(s, t)$ .
- (SC2) **Triple composition.** If a 3-vertex submosaic on  $(v_1, v_2, v_3)$  has labels  $(R_{12}, R_{13}, R_{23})$ , then  $R_{13} \in \text{comp}(R_{12}, R_{23})$ .
- (SC3) **Typed-EQ replacement (v2.1).** If  $v \in V(\mu)$  and  $v'$  is an endpoint summary with  $(\text{st}_d(v), \pi(v)) \equiv_{\text{EQ},Q} (\text{st}_d(v'), \pi(v'))$  (i.e.  $v$  and  $v'$  are linked by the explicit equality relation of Definition 10.1), then replacing  $v$  by  $v'$  in  $\mu$  yields a mosaic in  $\text{Mos}_k(s, t)$ . Replacement on alphabet coincidence alone is not licensed.
- (SC4) **One-step containment extension.** Adding a generated PPI-child of a vertex (with appropriate label propagation by composition with PP or PPI) yields a mosaic in  $\text{Mos}_k(s, t)$ .

**Lemma 8.7** (Support closure is finite). *For fixed rank  $k$ , the support closure of any finite set of mosaics is finite and effectively computable. By the mosaic-locality theorem of Section 12*

(Theorem 12.4, v2.1), it suffices to enumerate mosaics of arity at most  $N(C_0) = 3$ , plus the side-context arity bound 5 from Lemma 9.5. The total universe of relevant mosaics is bounded by  $|E_k|^5 \cdot |\{\text{DR}, \text{PO}, \text{EQ}, \text{PP}, \text{PPI}\}|^{\binom{5}{2}}$ , an effectively computable function of  $|\text{cl}(C_0)|$ .

*Proof.* The endpoint summary alphabet  $E_k$  is finite (Definition 8.4). By Theorem 12.4, every mosaic is determined (up to membership in the descriptor) by its triples; equivalently, a labelled complete graph on  $V$  satisfies (M1)–(M7) iff every 3-vertex sub-network does. Each closure operation in (SC1)–(SC4) preserves arity-3-locality. Iteration over the finite universe of arity- $\leq 5$  mosaics on  $E_k$ -labelled vertices terminates in effectively bounded time.  $\square$

### 8.3 Extraction from a model

**Construction 8.8** (Mosaic extraction). For each ordered child pair  $(s, t)$  in the split tree at rank  $k$ :

1. For every finite vertex set  $V \subseteq T_s \cup T_t \cup \{s, t\}$ , form the *realized mosaic*  $\mu_V$  with  $\mu_V(v, w) = \rho(\text{orig}(v), \text{orig}(w))$  and endpoint summaries given by the rank- $k$  profile, status, and equality-port tuple  $\pi(v)$  of each  $v \in V$ . The visible-equality predicate  $\equiv_{\text{EQ}, Q}^{\mu_V}$  is inherited from the canonical  $\equiv_{\text{EQ}, Q}$  of Lemma 10.11.
2. Initialize  $\text{Mos}_k^{\text{raw}}(s, t)$  as the set of realized mosaics for  $V$  ranging over finite subsets up to a bounded vertex count.
3. Close under (SC1)–(SC4) of Definition 8.6.

The result  $\text{Mos}_k(s, t)$  is the *extracted sibling descriptor*.

**Lemma 8.9** (Extracted descriptors are coherent). *The descriptor  $\text{Mos}_k(s, t)$  extracted in Construction 8.8 satisfies (M1)–(M7) and (SC1)–(SC4).*

*Proof.* (M1)–(M3) hold for realized mosaics because  $\mathcal{I}_w$  is a strong abstract RCC5 frame (Section 3); they are preserved under (SC1)–(SC4) closure operations. (M4) holds by Lemma 3.2. (M5)–(M6) hold by Lemma 8.2 and Lemma 8.3. (M7, v2.1) holds because, in the extracted mosaic,  $\mu(v, w) = \text{EQ}$  iff  $\text{orig}(v) = \text{orig}(w)$  in  $\mathcal{I}_w$ , which by the canonical port assignment of Lemma 10.11 holds iff  $(\text{st}_d(v), \pi(v)) \equiv_{\text{EQ}, Q} (\text{st}_d(w), \pi(w))$ . (SC1)–(SC4) hold by construction (we explicitly close).  $\square$

## 9 Witness menus for DR/PO existentials

For each rank- $d$  profile  $\sigma \in S$  and each  $\exists R.D \in \lambda_Q(\sigma)$  with  $R \in \{\text{DR}, \text{PO}\}$ , the witness menu records a target profile and relation.

**Construction 9.1** (Witness-menu extraction). For each  $u \in T$  with  $\text{st}_d(u) = \sigma$  and each  $\exists R.D \in \lambda(u)$  with  $R \in \{\text{DR}, \text{PO}\}$ :

1. Locate the selected witness  $b = w(\text{orig}(u), \exists R.D) \in D_w(a_0)$  from the witness policy.
2. Find a tree occurrence  $v$  with  $\text{orig}(v) = b$  (such  $v$  exists by Construction 6.1, possibly as a sibling occurrence under a common ancestor).

3. Record the witness-menu entry  $(\exists R.D) \mapsto (\text{st}_d(v), R)$  in  $\text{Wit}_d(\sigma)$ .

**Lemma 9.2** (Witness-menu relation-safety). *Every entry  $(\exists R.D) \mapsto (\sigma', R)$  in  $\text{Wit}_d(\sigma)$  satisfies:  $\text{Need}_R(\lambda_Q(\sigma)) \subseteq \lambda_Q(\sigma')$  and  $\text{Need}_{\text{inv}(R)}(\lambda_Q(\sigma')) \subseteq \lambda_Q(\sigma)$ .*

*Proof.* By Lemma 3.2 applied to the model edge  $(\text{orig}(u), \text{orig}(v))$  with relation  $R$ , and the fact that  $\lambda_Q(\sigma)$  depends only on  $\sigma$  (the closure type is the first profile component).  $\square$

*Remark 9.3.* PP and PPI existentials do *not* appear in the witness menu. They are discharged by reachability over the parent function of  $Q$  (Section 13). This separation is deliberate: PP/PPI witnesses are tracked structurally by the tree backbone, while DR/PO witnesses are tracked combinatorially by the menu.

### 9.1 Side-context witness lemma (v2.1, obligation O5)

Construction 9.1 places a DR/PO-witness occurrence  $v$  in the split tree, but the placement is described schematically (“possibly as a sibling occurrence under a common ancestor”). GPT-5.5’s review of v2 [8] flagged that completeness requires a formal lemma ensuring this placement preserves all relation-safety and composition constraints *at* the bounded interface where  $u$  and  $v$  meet. We supply that lemma here.

**Definition 9.4** (Side context of an occurrence pair). For occurrences  $u, v \in T$  with  $\text{orig}(u) = a$ ,  $\text{orig}(v) = b$  and  $\rho(a, b) \in \{\text{DR}, \text{PO}\}$ , the *side context* of the pair  $(u, v)$  in  $T$  is the finite vertex set

$$\text{Side}(u, v) := \{u, v, \text{par}(u), \text{par}(v), \text{lca}_T(u, v)\},$$

where  $\text{lca}_T(u, v) \in T$  is the (unique) lowest common ancestor of  $u$  and  $v$  in  $T$  if it exists, omitted otherwise; and where any of  $\text{par}(u)$ ,  $\text{par}(v)$  that fail to be defined (i.e. when  $u$  or  $v$  is the root of its tree) are also omitted. By construction,  $\text{Side}(u, v)$  has at most five elements:

- (i) the two endpoints  $u, v$  themselves (always present);
- (ii) the parent  $\text{par}(u)$  (one element, if  $u$  is not a root; otherwise zero);
- (iii) the parent  $\text{par}(v)$  (one element, if  $v$  is not a root; otherwise zero);
- (iv) the lowest common ancestor  $\text{lca}_T(u, v)$  (one element, if  $u$  and  $v$  lie in the same tree of the split forest; otherwise zero).

Counting:  $2 + 1 + 1 + 1 = 5$ , and any of the optional elements may also coincide with one of the others (e.g. if  $u$  and  $v$  are siblings then  $\text{lca}_T(u, v) = \text{par}(u) = \text{par}(v)$ , collapsing three of the five slots into one), which only further decreases  $|\text{Side}(u, v)|$ . Hence  $|\text{Side}(u, v)| \leq 5$  in all cases, and the bound is tight (it is attained when  $u$  and  $v$  have distinct, non-root parents and a strict common ancestor above both).

**Lemma 9.5** (Side-context witness lemma, v2.1). *Let  $u \in T$  be an occurrence with  $\text{orig}(u) = a$ , let  $\exists R.D \in \lambda(u)$  with  $R \in \{\text{DR}, \text{PO}\}$ , and let  $b = w(a, \exists R.D) \in D_w(a_0)$  be the selected witness from the witness policy. Then there exists an occurrence  $v \in T$  with  $\text{orig}(v) = b$  such that:*

- (W1)  $\rho(\text{orig}(u), \text{orig}(v)) = R$  (witness relation realized);

- (W2)  $D \in \lambda(v)$  (witness contains the body type);
- (W3) the realized mosaic  $\mu_{\text{Side}(u,v)}$  on the side context satisfies (M1)–(M7) of Definition 8.5 and is in the support closure  $\text{Mos}_k(\cdot, \cdot)$  for the relevant sibling pair (Construction 8.8);
- (W4) every universal  $\forall S.E \in \lambda(u)$  with  $S \in \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$  that fires on the realized  $u$ – $v$  edge is satisfied: if  $S = R$ , then  $E \in \lambda(v)$ ; symmetrically for  $\forall \text{inv}(S).E \in \lambda(v)$  targeting  $u$ ;
- (W5) every triple  $(u, v, z)$  with  $z \in \text{Side}(u, v)$  has  $\rho(\text{orig}(u), \text{orig}(z)) \in \text{comp}(R, \rho(\text{orig}(v), \text{orig}(z)))$  (RCC5 composition constraint at the interface).

*Proof.* Existence of  $v$ : by Construction 6.1 step 4 (sibling expansion), the split tree  $T$  contains an occurrence  $v$  with  $\text{orig}(v) = b$ . Either  $v$  is created *de novo* as a sibling under the lowest common ancestor of  $u$  and a tree node mapped to  $b$ , or  $v$  already exists in  $T$  as an occurrence of  $b$  generated earlier; both are licensed by step 4.

(W1). By the witness policy,  $\rho(a, b) = R$  in  $\mathcal{I}_w$ . Since the edge labelling  $\rho_T$  in  $T$  is pulled back from  $\rho$  via  $\text{orig}$  (Construction 6.1 step 6), we have  $\rho_T(u, v) = \rho(\text{orig}(u), \text{orig}(v)) = R$ .

(W2).  $b \in D_w(a_0)$  is a witness for  $\exists R.D$ , hence  $b \in D^{\mathcal{I}_w}$ , so  $D \in \text{tp}_{\mathcal{I}_w}(b) = \lambda(v)$ . (Witness selection guarantees the body type membership; this is a Hintikka-type closure property.)

(W3).  $\text{Side}(u, v)$  is a finite vertex set of size at most 5 (Definition 9.4). The realized mosaic  $\mu_{\text{Side}(u,v)}$  is constructed by setting  $\mu(v_1, v_2) = \rho(\text{orig}(v_1), \text{orig}(v_2))$  and equipping each vertex with its rank- $k$  profile, status, and equality-port tuple. Verifying (M1)–(M7):

- (M1)  $\mu(v_i, v_i) = \rho(\text{orig}(v_i), \text{orig}(v_i)) = \text{EQ}$  (RCC5 reflexivity).
- (M2) Inverse symmetry:  $\rho(b, a) = \text{inv}(\rho(a, b))$  in any RCC5 frame.
- (M3) Triple composition: every realized triple in  $\mathcal{I}_w$  satisfies the composition table by the strong abstract RCC5 frame property.
- (M4) Relation-safety: every realized model edge is relation-safe by Lemma 3.2.
- (M5)–(M6) Status-partition consequences from Lemma 8.2 and Lemma 8.3: the placement of  $v$  as a sibling under the LCA of  $u$  and  $v$  produces **Out** or **Front** status as determined by  $R \in \{\text{DR}, \text{PO}\}$ . If  $R = \text{DR}$ ,  $v$  has status **Out** relative to  $u$ ; (M5) gives only **DR** on cross-status edges from  $v$ , consistent with  $\rho(b, a) = \text{DR}$ . If  $R = \text{PO}$ ,  $v$  has status **Front**, and (**Front**, **Front**) cross-status edges have labels in  $\{\text{DR}, \text{PO}\}$  by (M6), consistent with  $\rho(b, a) = \text{PO}$ .
- (M7, v2.1) Visible-equality on  $\mu_{\text{Side}(u,v)}$  is by the canonical equality-port assignment of Lemma 10.11:  $\mu(v_1, v_2) = \text{EQ}$  iff  $\text{orig}(v_1) = \text{orig}(v_2)$  iff  $(\text{st}_d(v_1), \pi(v_1)) \equiv_{\text{EQ}, Q} (\text{st}_d(v_2), \pi(v_2))$ .

The mosaic  $\mu_{\text{Side}(u,v)}$  is therefore well-defined and satisfies (M1)–(M7). By Construction 8.8,  $\mu_{\text{Side}(u,v)}$  is added to  $\text{Mos}_k^{\text{raw}}(s, t)$  for the sibling pair  $(s, t)$  above  $u$  and  $v$ , then closed under (SC1)–(SC4). Thus  $\mu_{\text{Side}(u,v)} \in \text{Mos}_k(s, t)$ .

(W4).  $\mathcal{I}_w$  satisfies all universals in  $C_0$  (Lemma 4.3). In particular, if  $\forall S.E \in \lambda(u) = \text{tp}_{\mathcal{I}_w}(a)$  and  $\rho(a, b) = S$ , then  $E \in \text{tp}_{\mathcal{I}_w}(b) = \lambda(v)$ . Symmetrically for  $\forall \text{inv}(S).E \in \lambda(v)$ . When  $S = R$ , this is precisely the universal-firing condition.

(W5). For any  $z \in \text{Side}(u, v)$ , the realized triple  $(\text{orig}(u), \text{orig}(v), \text{orig}(z))$  in  $\mathcal{I}_w$  satisfies the

composition constraint of the strong abstract RCC5 frame:

$$\rho(\text{orig}(u), \text{orig}(z)) \in \text{comp}(\rho(\text{orig}(u), \text{orig}(v)), \rho(\text{orig}(v), \text{orig}(z))) = \text{comp}(R, \rho(\text{orig}(v), \text{orig}(z))).$$

Since the mosaic  $\mu_{\text{Side}(u,v)}$  records the realized relations faithfully, this composition is captured in the mosaic’s edge labels, satisfying (M3) for the triple.  $\square$

*Remark 9.6* (Why a *bounded* side context suffices). The side context  $\text{Side}(u, v)$  has at most 5 vertices, regardless of where  $u$  and  $v$  sit in  $T$ . This boundedness is essential for the finite-index property: the support-closed mosaic  $\text{Mos}_k(s, t)$  contains only finitely many mosaics of bounded arity (Lemma 8.7), so the witness menu’s interface data is finitely representable.

The lemma’s bound of 5 is the maximum needed for the immediate  $u$ – $v$  interface together with one-step ancestors and the LCA; deeper composition constraints involving  $u$  or  $v$  are captured by support-closure operations (SC1)–(SC4), not by extending the side context itself.

*Remark 9.7* (Relation to obligation O5). GPT-5.5’s recommendation [8, §Side-context witness lemma] called for a lemma ensuring that DR/PO witnesses placed by the witness policy are realized in the split presentation while preserving relation-safety, universal restrictions, RCC5 composition constraints, and consistency with sibling mosaics and typed-EQ mate data. Lemma 9.5 discharges this obligation by the bookkeeping argument above: every witness occurrence sits in a bounded side context, the realized mosaic on that side context satisfies all (M1)–(M7) axioms, and the universals/compositions involving the interface are inherited from the validity of  $\mathcal{I}_w$  as a strong-EQ model of  $C_0$ .

**Caveat.** The lemma is a *local* statement about a single witness pair. The lift to a global statement about *all* witness occurrences simultaneously is the coherent eventuality realization obligation (O6), discharged in Section 14 (Lemma 14.6).

## 10 Typed-EQ congruence (v2.1: explicit equality data)

**(v2.1, obligation O1.)** GPT-5.5’s review of v2 [8] flagged a tension between two earlier statements:

- *Profiles may legitimately repeat on a blocking loop* (Remark 7.4: distinct unfolding occurrences with the same profile are not EQ-related).
- *Typed-EQ colour was a profile component* (Definition 7.1, item 2; mosaic axiom (M7): “same colour implies EQ-label”).

A reusable colour on a blocking lap could collapse an infinite PP-chain into semantic EQ. The fix below makes the equality data *explicit* and *local* to the certificate, so profile equality alone never implies  $\equiv_{\text{EQ}}$ .

### 10.1 Explicit equality ports

**Definition 10.1** (Equality ports and finite equality relation). A *rank- $d$  equality port* on a profile  $\sigma \in S$  is a formal label  $p$  drawn from a finite *equality alphabet*  $\Pi_\sigma$ . An occurrence  $u \in T$



carries a finite tuple of ports  $\pi(u) = (\pi_1(u), \dots, \pi_n(u))$  with  $\pi_i(u) \in \Pi_{\text{st}_d(u)}$ . The certificate  $Q$  is augmented with an *explicit finite equality relation*

$$\equiv_{\text{EQ}, Q} \subseteq \bigsqcup_{\sigma, \sigma' \in S} (\{\sigma\} \times \Pi_\sigma) \times (\{\sigma'\} \times \Pi_{\sigma'})$$

declared in advance and stored as part of the finite quotient.  $(\sigma, p) \equiv_{\text{EQ}, Q} (\sigma', q)$  means: *the certificate asserts that an occurrence of  $\sigma$  carrying port  $p$  is the same semantic element as an occurrence of  $\sigma'$  carrying port  $q$ .*

**Definition 10.2** (Visible-equality predicate on a mosaic). Let  $\mu$  be an atomic mosaic with vertex set  $V(\mu)$ . The *visible equality* on  $V(\mu)$  is

$$v \equiv_{\text{EQ}, Q}^\mu w :\iff (\text{st}_d(v), \pi(v)) \equiv_{\text{EQ}, Q} (\text{st}_d(w), \pi(w)).$$

Equality data is local to the mosaic: only port identifiers appearing as endpoint summaries on  $V(\mu)$  are relevant when checking  $\equiv_{\text{EQ}, Q}^\mu$ .

## 10.2 Revised mosaic axiom (M7)

(M7) [v2.1, replaces v2 (M7)]. For every mosaic  $\mu$  and vertices  $v, w \in V(\mu)$ :

$$\mu(v, w) = \text{EQ} \iff v \equiv_{\text{EQ}, Q}^\mu w.$$

That is: the EQ-label appears on a mosaic edge *iff* the two endpoints are linked by the explicit equality relation  $\equiv_{\text{EQ}, Q}$  visible in that mosaic.

**What changed.** v2's (M7) read “endpoints with the same typed-EQ colour have label EQ”. Under v2's profile-component formulation, this could fire for any two occurrences whose profiles coincided on a blocking loop, even with no semantic EQ relation. v2.1's (M7) requires that equality be *declared* in  $\equiv_{\text{EQ}, Q}$ , not merely observed in profile equality. Profile equality is now *necessary but not sufficient* for the EQ-label.

## 10.3 Vertical separation axiom

**Definition 10.3** (Vertical separation, v2.1). The certificate  $Q$  satisfies the *vertical-separation axiom* when, for all profiles  $\sigma, \sigma' \in S$  and all ports  $p \in \Pi_\sigma, q \in \Pi_{\sigma'}$ :

$$\sigma \rightsquigarrow_{\text{PP}} \sigma' \text{ or } \sigma \rightsquigarrow_{\text{PPI}} \sigma' \quad (\text{strict, i.e. via at least one parent step}) \implies (\sigma, p) \not\equiv_{\text{EQ}, Q} (\sigma', q).$$

That is: any two profile-port pairs connected by a strict vertical PP/PPI-path can never be EQ-identified.

*Remark 10.4* (Why this is the right rule). A strict PP or PPI step in  $\mathcal{I}_w$  is irreflexive (RCC5 frame requirement on  $\{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$ ), so  $a \text{ PP } b \Rightarrow a \neq b$ . Hence semantic EQ-mates can never lie on a strict vertical chain in any model. The vertical-separation axiom imports this fact into the finite certificate and rules out the bad case identified by GPT-5.5: a self-loop  $\text{par}_Q(\sigma) = \sigma$  with



$(\sigma, p) \equiv_{\text{EQ}, Q} (\sigma, p)$  would force same-element identification across a loop traversal that represents a strict PP step in the unfolding. The axiom forbids that.

#### 10.4 Blocking equivalence is now distinct from equality

**Definition 10.5** (Blocking equivalence on the unfolding). On the unfolding of  $Q$ , two occurrences  $u, v$  are *blocking-equivalent*, written  $u \equiv_{\text{blk}} v$ , iff  $\text{st}_d(u) = \text{st}_d(v)$  and  $\pi(u) = \pi(v)$  as port tuples. Blocking equivalence is a finite indexing device.

**Definition 10.6** (Semantic EQ-mate relation on the unfolding). On the unfolding of  $Q$ , two occurrences  $u, v$  are *semantic EQ-mates*, written  $u \equiv_{\text{EQ}} v$ , iff their profile-port pairs are linked by  $\equiv_{\text{EQ}, Q}$ :  $(\text{st}_d(u), \pi(u)) \equiv_{\text{EQ}, Q} (\text{st}_d(v), \pi(v))$ .

**Lemma 10.7** (Blocking does not imply equality, v2.1). *Under the vertical-separation axiom,  $u \equiv_{\text{blk}} v$  does not imply  $u \equiv_{\text{EQ}} v$ . In particular, distinct unfolding occurrences on a blocking self-loop are blocking-equivalent but not semantic EQ-mates.*

*Proof.* Distinct occurrences on a self-loop are connected by a strict PP-path of length  $\geq 1$  in the unfolding. By Definition 10.3,  $(\sigma, p) \not\equiv_{\text{EQ}, Q} (\sigma, p)$  when the same  $(\sigma, p)$  pair is connected by a strict PP-path. Hence  $u \equiv_{\text{EQ}} v$  fails even when  $u \equiv_{\text{blk}} v$  holds.  $\square$

**Definition 10.8** (Mate class of a profile, v2.1). For  $\sigma \in S$ , the *mate class*

$$\text{Mate}(\sigma) := \{\sigma' \in S : \exists p \in \Pi_\sigma, q \in \Pi_{\sigma'} \text{ with } (\sigma, p) \equiv_{\text{EQ}, Q} (\sigma', q)\} \cup \{\sigma\}.$$

This is the set of profiles that some equality-port assignment on  $\sigma$  may declare as semantically equal to  $\sigma$ .

#### 10.5 Profile component (item 2) is replaced

The original Definition 7.1 listed “typed-EQ colour  $c_{\text{EQ}}(u)$ ” as profile component 2. Under v2.1 this is replaced by “equality-port tuple  $\pi(u)$ ”. The finite-index argument (Lemma 7.2) goes through unchanged: ports are drawn from a finite alphabet whose size is bounded by the number of distinct semantic elements that can appear in a witness-generated submodel cell, hence by an effectively computable function of  $|\text{cl}(C_0)|$ .

**Definition 10.9** (Typed-EQ congruence, v2.1). A typed-EQ congruence on  $T$  is an equivalence relation  $\equiv_{\text{EQ}}$  such that for all  $x \equiv_{\text{EQ}} y$ :

- (EQ1)  $\lambda(x) = \lambda(y)$  (same closure type);
- (EQ2)  $(\text{st}_d(x), \pi(x)) \equiv_{\text{EQ}, Q} (\text{st}_d(y), \pi(y))$  (linked by the explicit equality relation);
- (EQ3) the vertical eventuality summaries at  $x$  and  $y$  agree;
- (EQ4) for every other equivalence class  $[z]$ , the relation  $\rho(\text{orig}(x), \text{orig}(z'))$  is the same for all  $z' \in [z]$  and matches  $\rho(\text{orig}(y), \text{orig}(z'))$ ;
- (EQ5)  $x$  and  $y$  are not connected by a strict PP/PPI-path in the unfolding (vertical separation, automatic from (EQ2) combined with Definition 10.3).

*Remark 10.10* (Blocking versus equality, v2.1). Lemma 10.7 restates the central separation:  $\equiv_{\text{blk}}$  (finite blocking) and  $\equiv_{\text{EQ}}$  (semantic EQ) are now distinct relations. Profile equality, or equality of any profile component including the equality-port *alphabet*, never by itself implies  $\equiv_{\text{EQ}}$ ; only an explicit  $\equiv_{\text{EQ},Q}$  entry does.

**Lemma 10.11** (Extracted congruence, v2.1). *Let  $u, v$  be occurrences in the split tree  $T$  of Construction 6.1. Define  $u \equiv_{\text{EQ}} v$  iff  $\text{orig}(u) = \text{orig}(v)$  in  $\mathcal{I}_w$ , and define ports  $\pi(u)$  canonically as the index of  $\text{orig}(u)$  in a fixed enumeration of the EQ-mate class of  $\text{orig}(u)$  within  $\mathcal{I}_w$ . Then:*

1. *The induced  $\equiv_{\text{EQ},Q}$  on profile-port pairs is well-defined and finite.*
2. *The vertical-separation axiom holds: if  $\sigma \rightsquigarrow_{\text{PP}} \sigma'$  strictly, no port assignment links  $(\sigma, p) \equiv_{\text{EQ},Q} (\sigma', q)$ .*
3.  *$\equiv_{\text{EQ}}$  on  $T$  satisfies (EQ1)–(EQ5) of the v2.1 Definition 10.9.*

*Proof.* (1) Under strong EQ, every semantic element is its own EQ-class, so the canonical port assignment is the identity index; the  $\equiv_{\text{EQ},Q}$  relation is trivial:  $(\sigma, p) \equiv_{\text{EQ},Q} (\sigma', q)$  iff  $p = q$  corresponds to the same semantic element appearing in two profile cells. Finiteness: an element of  $\mathcal{I}_w$  contributes at most one port per profile cell it appears in, and there are at most  $|S|$  profile cells per element; total ports per element are bounded by  $|S|$ .

(2) A strict PP-path in the unfolding lifts to a strict PP-chain in  $\mathcal{I}_w$  on the origin elements. RCC5 irreflexivity forbids  $aPPa$ , so endpoints of a strict PP-chain are distinct semantic elements; their canonical port identifiers therefore differ as well. Hence  $(\sigma, p) \not\equiv_{\text{EQ},Q} (\sigma', q)$  along strict vertical paths.

(3) (EQ1) is immediate. (EQ2) holds by construction of the canonical port assignment. (EQ3)–(EQ4) hold by the same arguments as in v2 (vertical summaries and cross-class relations are orig-invariant). (EQ5) is (2).  $\square$

## 11 Local coherence (C1)–(C5)

We verify the (C1)–(C5) coherence conditions of [5, Definition 1.8], now restated for mosaics.

**Theorem 11.1** (Local coherence). *The descriptors  $\text{Mos}_k(s, t)$  extracted in Construction 8.8 satisfy local coherence (C1)–(C5).*

*Proof.* **(C1) Out rigidity.** By (M5) of Definition 8.5, every Out-status endpoint contributes only DR-labelled cross-status edges.

**(C2) Front–Front type-safety.** Let  $\mu \in \text{Mos}_k(s, t)$  contain endpoints  $v_1, v_2$  with (Front, Front) status. By (M4),  $\mu(v_1, v_2) \in \{\text{DR}, \text{PO}\}$  is relation-safe for the closure types  $(\lambda(v_1), \lambda(v_2))$ .

**(C3) Left downward support.** Suppose  $\mu \in \text{Mos}_k(s, t)$  contains  $v_1$  on the  $s$ -side, and let  $v'_1$  be a generated PPI-child of  $\text{orig}(v_1)$  (equivalently, a child occurrence in  $T_s$ ). By (SC4) (one-step containment extension), the descriptor  $\text{Mos}_k(s, t)$  contains a mosaic  $\mu'$  extending  $\mu$  with  $v'_1$  added. The label  $\mu'(v'_1, v_2)$  lies in  $\text{comp}(\text{PPI}, \mu(v_1, v_2))$ , hence the required composition rule holds.

**(C4) Right downward support.** Symmetric to (C3) on the  $t$ -side.

**(C5) Triple coherence.** This is where v1’s witness conflation occurred. Let  $v_1, v_2, v_3 \in V(\mu)$  be three vertices in a mosaic  $\mu \in \text{Mos}_k(s, t)$  with the relevant statuses. Because  $\mu$  is an *atomic mosaic* (a joint network on  $V$ , not a collection of independent pair labels), it directly satisfies  $\mu(v_1, v_3) \in \text{comp}(\mu(v_1, v_2), \mu(v_2, v_3))$  by (M3).

The crucial point: the three relations  $\mu(v_1, v_2), \mu(v_2, v_3), \mu(v_1, v_3)$  are realized *by a single triple of witnesses* ( $\text{orig}(v_1), \text{orig}(v_2), \text{orig}(v_3)$ ) in  $\mathcal{I}_w$ , not by three independent pairs. This is enforced by Construction 8.8: a realized mosaic comes from a single vertex set  $V$ , with all pairs labelled by the *same* model.

Support closure (SC1)–(SC4) propagates this joint realizability: restriction preserves it (a sub-triple of a real triple is a real triple); typed-EQ replacement preserves it (an EQ-mate is realized by the same model element); one-step extension preserves it (composition through the model).  $\square$

*Remark 11.2* (How v1’s bug is repaired). v1 wrote “ $M((\chi_1, \chi_2))$  contains  $R_{12}$  from witnesses  $(x_1, x_2)$ ,  $M((\chi_1, \chi_3))$  contains  $R_{13}$  from witnesses  $(x'_1, x_3)$ , hence  $\rho(\text{orig}(x_2), \text{orig}(x_3)) \in \text{comp}(\text{inv}(R_{12}), R_{13})$  appears in  $M((\chi_2, \chi_3))$ .” The bug:  $x_1$  and  $x'_1$  may be distinct occurrences with the same profile but different concrete witnesses; the composition does *not* apply across them.

The repair: mosaics record *joint* networks on a single vertex set. Triple coherence is automatic because the three relations are realized by one triple of model elements, by construction.

## 12 Finite mosaic arity bound (v2.1, obligation O4)

GPT-5.5’s v2 review ([8], obligation O4) flagged that the v2 statement “mosaics may be bounded by  $|\text{cl}(C_0)| + 4$  vertices because larger mosaics decompose into overlapping smaller ones” is informal: a rigorous treatment requires a *locality theorem* establishing an effective bound  $N(C_0)$  such that all mosaic axioms, support-closure operations, and witness checks are local below  $N(C_0)$ .

This section discharges that obligation by proving 3-*locality* of the mosaic axioms (M1)–(M7) and 3-*locality* of (SC2) triple composition, together with 5-*locality* of the side-context witness lemma. The certificate’s enumerable arity is therefore  $N(C_0) = 5$  (a *constant*, not a function of  $|\text{cl}(C_0)|$ ); the dependence of the certificate’s size on  $C_0$  enters only through the finite endpoint-summary alphabet  $E_k$ .

### 12.1 Locality of the mosaic axioms

**Lemma 12.1** (Arity bound on individual mosaic axioms). *Each axiom (M1)–(M7) of Definition 8.5 is satisfied by a labelled complete graph  $\mu$  iff it is satisfied by every sub-network of  $\mu$  on at most 3 vertices.*

*Proof.* **(M1)**  $\mu(v, v) = \text{EQ}$  is a self-loop check at arity 1. **(M2)**  $\mu(w, v) = \text{inv}(\mu(v, w))$  is a pairwise check at arity 2. **(M3)**  $\mu(u, w) \in \text{comp}(\mu(u, v), \mu(v, w))$  is the RCC5 composition check at arity 3. **(M4)** Edge relation-safety (Definition 3.1) is a pairwise check at arity 2:  $\text{Need}_{\mu(v, w)}(\lambda(v)) \subseteq \lambda(w)$  and the dual. **(M5)** The Out-rigidity axiom is a pairwise check at

arity 2: every Out-status endpoint contributes a single label DR. **(M6)** The Front–Front open-cross axiom is a pairwise check at arity 2: every (Front, Front) pair carries a label in {DR, PO}. **(M7, v2.1)** The  $\equiv_{\text{EQ},Q}$ -coincidence axiom is a pairwise check at arity 2:  $\mu(v, w) = \text{EQ}$  iff  $(\text{st}_d(v), \pi(v)) \equiv_{\text{EQ},Q} (\text{st}_d(w), \pi(w))$ .

The maximum arity required is 3, achieved by (M3). Each axiom quantifies universally over all  $V$ -tuples; satisfaction at arity  $\leq 3$  on every sub-tuple suffices.  $\square$

## 12.2 Locality of the support-closure operations

**Lemma 12.2** (Arity bound on support-closure operations). *The support-closure operations (SC1)–(SC4) of Definition 8.6 all act locally below arity 5:*

- (i) (SC1) *Restriction is monotone-decreasing in arity.*
- (ii) (SC2) *Triple composition is arity 3.*
- (iii) (SC3) *Typed-EQ replacement is arity 2 (one-vertex substitution; pairwise relations to other vertices are preserved by definition of  $\equiv_{\text{EQ},Q}$ ).*
- (iv) (SC4) *One-step containment extension adds at most one new vertex, raising arity by 1. Combined with the side-context witness lemma 9.5, the total active arity at any extension step is  $\leq 5$ .*

*Proof.* (SC1)–(SC3) are immediate from the cited definitions.

For (SC4): the extension takes a mosaic  $\mu$  over vertex set  $V$  and adds a new vertex  $v'$  corresponding to a generated PPI-child of some  $v \in V$ , with the new edges  $\mu'(v', w) := \rho(\text{orig}(v'), \text{orig}(w))$  for  $w \in V$ . By Lemma 9.5, the relevant interaction involves only the side context  $\text{Side}(v', w)$  of size at most 5, not the full mosaic. Hence the local check is 5-local.  $\square$

## 12.3 Patchwork-based locality theorem

For self-containment we restate the path-consistency theorem of Renz–Nebel [1] in the precise form we use.

**Fact 12.3** (Renz–Nebel path-consistency theorem, atomic case). Let  $\mathcal{N} = (V, \rho)$  be a labelled directed graph with finite vertex set  $V$  and labels  $\rho(x, y) \in \{\text{DR}, \text{PO}, \text{EQ}, \text{PP}, \text{PPI}\} = \{\text{DR}, \text{PO}, \text{EQ}, \text{PP}, \text{PPI}\}$  for every ordered pair  $x \neq y$ , satisfying:

- (i) (*Reflexivity*)  $\rho(x, x) = \text{EQ}$ ;
- (ii) (*Converse closure*)  $\rho(y, x) = \text{inv}(\rho(x, y))$ , where  $\text{inv}$  swaps  $\text{PP} \leftrightarrow \text{PPI}$  and fixes  $\text{DR}, \text{PO}, \text{EQ}$ ;
- (iii) (*Triangle composition support*) for every triple of distinct vertices  $x, y, z$ ,

$$\rho(x, z) \in \text{comp}(\rho(x, y), \rho(y, z)),$$

where  $\text{comp}$  is the RCC5 composition table.

Then there exists a topological model  $(\Delta, \text{interp})$  with concrete RCC5 regions  $r_x \subseteq \Delta$  for  $x \in V$  realizing every labelled edge:  $\rho(x, y) = R$  implies  $r_x$  and  $r_y$  stand in relation  $R$  in  $\Delta$ .

The crucial consequence for us is that *global realizability of an atomic RCC5 network reduces to the 3-local condition (iii) above*. Equivalently: if every triangle of  $\mathcal{N}$  is in the support of composition, then  $\mathcal{N}$  is realizable by a concrete frame. This is the content the patchwork property contributes to mosaic locality.

**Theorem 12.4** (Mosaic locality, v2.1). *There is an effectively computable constant  $N(C_0) = 5$  such that for every concept  $C_0$  and every rank  $k \leq d(C_0)$ :*

- (a) *every mosaic axiom (M1)–(M7) is 3-local (Lemma 12.1);*
- (b) *every support-closure operation (SC1)–(SC4) is 5-local (Lemma 12.2);*
- (c) *the side-context witness lemma is 5-local (Lemma 9.5, condition  $|\text{Side}| \leq 5$ );*
- (d) *by the patchwork property of RCC5 [1], every labelled complete graph  $\mu$  on endpoint summaries from  $E_k$  belongs to the support-closed descriptor  $\text{Mos}_k(s, t)$  iff every sub-network of  $\mu$  of arity  $\leq 5$  does.*

*Proof.* Parts (a)–(c) restate Lemmas 12.1, 12.2, and 9.5 (the size- $\leq 5$  clause).

For (d): a labelled complete graph  $\mu$  over a finite vertex set  $V$  defines a binary RCC5 relation network in the sense of Renz–Nebel [1]. By Fact 12.3,  $\mu$  (in atomic form) is realizable by a concrete RCC5 frame iff every triangle of  $\mu$  is in the support of RCC5 composition. Triangle support is exactly 3-locality of (M3) and (SC2).

For atomic mosaics (Definition 8.5), every edge is a single base relation (not a disjunction), so path-consistency on triangles equals exact composition in the RCC5 table. Triangle satisfaction of (M1)–(M7) is exactly 3-locality of those axioms. Triangle satisfaction of (SC2) is exactly the composition condition. The remaining closure operations (SC1, SC3, SC4) preserve 3-locality (as shown in Lemma 12.2); (SC4) in addition uses 5-locality from the side-context witness lemma.

Combining: a labelled complete graph satisfies (M1)–(M7) and is support-closed iff every  $\leq 5$ -vertex sub-network satisfies these conditions. Hence  $N(C_0) = 5$  suffices.  $\square$

**Corollary 12.5** (Effective enumerability of mosaics). *For any concept  $C_0$ , the set of support-closed mosaic descriptors is effectively enumerable in time bounded by  $|E_d|^{5 \cdot |\{\text{DR, PO, EQ, PP, PPI}\}|^{(5)}}$ , where  $|E_d| = |\Sigma_d| \cdot 3 \cdot |\Pi_{\Sigma_d}|$  is the size of the endpoint-summary alphabet at the maximum rank.*

*Proof.* Enumerate all labelled complete graphs on at most 5 vertices over  $E_d$ , filter by (M1)–(M7) and the closure axioms. By Theorem 12.4, the resulting set is the support-closed descriptor.  $|\Sigma_d|$  is finite by Lemma 7.2;  $|\Pi_{\Sigma_d}|$  is finite by Lemma 10.11 (each port alphabet has size bounded by the number of EQ-mate slots);  $|\{\text{DR, PO, EQ, PP, PPI}\}| = 5$ .  $\square$

*Remark 12.6* (Comparison with the v2 informal bound). The v2 manuscript informally bounded mosaic arity by  $|\text{cl}(C_0)| + 4$ , claiming that “larger mosaics decompose into overlapping smaller ones via restriction.” This is now replaced by the explicit constant  $N(C_0) = 5$ , decoupled from  $|\text{cl}(C_0)|$ . The dependence on  $C_0$  enters only through the endpoint-summary alphabet  $|E_d|$ , which appears polynomially in the enumeration cost (Corollary 12.5).  $N(C_0)$  is constant because (M1)–(M7) and (SC1)–(SC4) are all binary or ternary, and the side-context witness lemma is 5-bounded.

*Remark 12.7* (O4 discharged). Theorem 12.4 discharges GPT-5.5's obligation O4 [8]. The proof routes: (i) axioms (M1)–(M7) are  $\leq 3$ -local by Lemma 12.1; (ii) closure operations (SC1)–(SC4) are  $\leq 5$ -local by Lemma 12.2; (iii) atomic RCC5 networks are triangle-locally consistent (Renz–Nebel [1]); (iv) the side-context witness lemma 9.5 fits within the 5-arity bound. The certificate enumeration is therefore controlled by  $|E_d|^5$ , which is an effectively computable function of  $|\text{cl}(C_0)|$  via Lemma 7.2.

### 13 Reachability discharge for PP/PPI eventualities

This section provides the finitary alternative to GPT-5.5's cycle-realization condition [6, Definition 6.4].

**Definition 13.1** (Parent reachability in  $Q$ ). For  $\sigma, \sigma' \in S$ , write  $\sigma \rightsquigarrow_{\text{PP}} \sigma'$  if there exist  $\sigma_0, \sigma_1, \dots, \sigma_n$  in  $S$  with  $\sigma_0 = \sigma$ ,  $\sigma_n = \sigma'$ ,  $n \geq 1$ , and  $\text{par}_Q(\sigma_i) = \sigma_{i+1}$  for  $0 \leq i < n$ . Self-loops are permitted:  $\text{par}_Q(\sigma) = \sigma$  implies  $\sigma \rightsquigarrow_{\text{PP}} \sigma$ .

Symmetrically,  $\sigma \rightsquigarrow_{\text{PPI}} \sigma'$  via the inverse parent function (i.e., child function), allowing self-loops.

**(v2.1, obligation O2.)** GPT-5.5's review of v2 [8] flagged that the v2 reachability statement quantified over a single occurrence  $u$  with  $\text{st}_d(u) = \sigma$ , but in a generated split tree distinct occurrences of the same profile  $\sigma$  may carry distinct port tuples, and only the union over all EQ-mate-equivalent occurrences is canonical. Under v2.1, both the existential discharge and the universal propagation are stated over the mate class  $\text{Mate}(\sigma)$  of Definition 10.8.

**Lemma 13.2** (Reachability discharge, v2.1). *Let  $Q$  be the quotient extracted in Construction 6.1 from a witness-generated model  $(\mathcal{I}_w, a_0)$ . For every  $\sigma \in S$  and every  $\exists R.D \in \lambda_Q(\sigma)$  with  $R \in \{\text{PP}, \text{PPI}\}$ ,*

$$\exists \hat{\sigma} \in \text{Mate}(\sigma) \exists \sigma' \in S \quad \hat{\sigma} \rightsquigarrow_R \sigma' \text{ and } D \in \lambda_Q(\sigma').$$

*That is: some mate of  $\sigma$  has an  $R$ -reachable target profile carrying  $D$ . Equivalently, the reachability check is run not from a single profile node  $\sigma$  but from the mate class  $\text{Mate}(\sigma) \subseteq S$ , treating the latter as a single source.*

*Proof.* We treat  $R = \text{PP}$ ;  $R = \text{PPI}$  is symmetric.

Let  $\sigma \in S$  with  $\exists \text{PP}.D \in \lambda_Q(\sigma)$ , and let  $u \in T$  be *any* occurrence with  $\text{st}_d(u) = \sigma$  (such  $u$  exists by definition of  $S$ ). By the closure of types under  $\equiv_{\text{EQ}}$  (Definition 10.9 (EQ1)), every semantic EQ-mate of  $\text{orig}(u)$  in  $\mathcal{I}_w$  has the same closure type as  $\text{orig}(u)$ , hence also satisfies  $\exists \text{PP}.D$ .

Choose a representative  $u^\dagger \in T$  at which the witness for  $\exists \text{PP}.D$  has actually been generated by Construction 6.1 step 2 (upward expansion);  $u^\dagger$  maps under  $\text{orig}$  to a semantic EQ-mate of  $\text{orig}(u)$ , hence the profile  $\hat{\sigma} := \text{st}_d(u^\dagger)$  lies in  $\text{Mate}(\sigma)$  (possibly  $\hat{\sigma} = \sigma$  itself). At  $u^\dagger$ , the parent chain reaches a finite occurrence  $u^\dagger = u_0, u_1, \dots, u_m$  with  $\text{par}(u_i) = u_{i+1}$  and  $D \in \lambda(u_m)$ . Projecting via  $\text{st}_d$ :

$$\hat{\sigma} = \text{st}_d(u_0) \xrightarrow{\text{par}_Q} \text{st}_d(u_1) \xrightarrow{\text{par}_Q} \dots \xrightarrow{\text{par}_Q} \text{st}_d(u_m),$$

giving  $\hat{\sigma} \rightsquigarrow_{\text{PP}} \sigma'$  with  $\sigma' = \text{st}_d(u_m)$  and  $D \in \lambda_Q(\sigma')$ .  $\square$



*Remark 13.3* (Why mate-quantified discharge is the right statement). In a generated split tree, the witness selector  $w$  may pick a witness for  $\exists \text{PP}.D$  at one occurrence  $u^\dagger$  of an EQ-mate class but not at every occurrence  $u$  of the same class. After the typed-EQ congruence is taken (and after splitting introduces multiple tree occurrences per semantic element), the quotient profile  $\sigma$  may be witnessed at several  $\hat{\sigma} \in \text{Mate}(\sigma)$ , and the canonical discharge is via the mate class rather than via the single literal profile  $\sigma$ . This was implicit in the v2 proof but is made explicit in v2.1.

When  $\text{Mate}(\sigma) = \{\sigma\}$  (no nontrivial  $\equiv_{\text{EQ},Q}$ -links to other profiles), the v2 statement and the v2.1 statement coincide.

*Remark 13.4* (This is finite-graph reachability, not Büchi). Lemma 13.2 reduces the discharge of transitive PP/PPI eventualities to ordinary reachability over the finite directed graph  $(S, \text{par}_Q)$ , decidable in  $O(|S| + |E_Q|)$  time per eventuality (e.g., breadth-first search).

GPT-5.5’s Definition 6.4 phrases the condition in terms of stem- and-cycle decompositions of branches, with realization required on the cycle. When the branch is summarized by a finite quotient  $Q$ , “cycle in the unfolding” becomes “cycle in  $\text{par}_Q$ ”. Realization on the cycle then becomes existence of a profile in the strongly connected component containing  $\sigma$  that has  $D$  in its type.

Restated as graph reachability:  $\exists \text{PP}.D \in \lambda_Q(\sigma)$  holds iff  $D$  appears in some profile reachable from  $\sigma$  via  $\text{par}_Q$ . This is decidable in polynomial time without Büchi acceptance,  $\omega$ -words, or any automata-theoretic machinery.

**Lemma 13.5** (Universal safety propagation, v2.1). *For every  $\sigma \in S$  and every  $\forall R.D \in \lambda_Q(\sigma)$  with  $R \in \{\text{PP}, \text{PPI}\}$ :*

$$\forall \hat{\sigma} \in \text{Mate}(\sigma) \forall \sigma' \in S \quad \hat{\sigma} \rightsquigarrow_R \sigma' \implies D \in \lambda_Q(\sigma').$$

*That is: along reachability from any mate of  $\sigma$ , the target profile carries  $D$  in its type.*

*Proof.* By (EQ1) of Definition 10.9, all profiles in  $\text{Mate}(\sigma)$  share the same closure type, hence  $\forall R.D \in \lambda_Q(\hat{\sigma})$  for every  $\hat{\sigma} \in \text{Mate}(\sigma)$ . Lemma 3.2 applied along the parent chain in  $\mathcal{I}_w$  then ensures every  $R$ -reachable target type contains  $D$ . Profile equality preserves type, so  $\lambda_Q(\sigma') \ni D$ .  $\square$

## 14 Simultaneous eventuality realization (v2.1, obligation O6)

Lemma 13.2 discharges each transitive eventuality *individually*: for each  $\exists R.D \in \lambda_Q(\sigma)$  separately, there is a mate-class source from which  $D$  is reachable in  $(S, \text{par}_Q)$ . GPT-5.5’s v2 review ([8], obligation O6) flagged that the validity argument requires more: a single *coherent infinite unfolding* in which *all* pending eventualities at *all* occurrences are realized *simultaneously*.

This section discharges that obligation for the extracted quotient. The key observation is that the split tree  $T$  itself (Construction 6.1) is already a concrete realization of the canonical unfolding of  $Q$ , with every existential witnessed by an actual occurrence by construction of the witness-generated submodel  $\mathcal{I}_w$ .

**Definition 14.1** (Canonical unfolding of  $Q$ ). The *canonical unfolding*  $U(Q)$  of the finite quotient  $Q$  is the rooted tree whose nodes are finite walks in  $(S, \text{par}_Q)$  starting at  $\sigma_*$ , with the parent map sending a walk to its one-step predecessor and the child relation determined by the profile component  $\text{Child}_Q$  (item 4 of Definition 7.1): a profile  $\sigma$  with multiplicity  $m(\sigma') \in \{1, 2+\}$  for child profile  $\sigma'$  contributes (at least)  $m(\sigma')$  child occurrences of profile  $\sigma'$ . Equality-port data of  $Q$  ( $\Pi_Q, \equiv_{\text{EQ}, Q}$ ) descends to a quotient on  $U(Q)$  identifying EQ-mate occurrences.

**Definition 14.2** (Projection  $\pi : T \rightarrow U(Q)$ ). For each occurrence  $u \in T$ , let  $\pi(u) \in U(Q)$  be the walk

$$\pi(u) := (\text{st}_d(u_*), \text{st}_d(\text{par}^{m-1}(u)), \dots, \text{st}_d(\text{par}(u)), \text{st}_d(u)),$$

where  $m$  is the depth of  $u$  in  $T$  relative to the root  $u_*$ . By Definition 7.3,  $\text{par}_Q \circ \text{st}_d = \text{st}_d \circ \text{par}$  on  $T$ , so this is a well-defined walk in  $(S, \text{par}_Q)$ .

**Lemma 14.3** (Projection  $\pi$  is a faithful unfolding map). *The map  $\pi : T \rightarrow U(Q)$  of Definition 14.2 satisfies:*

- (i)  $\pi$  is parent-respecting:  $\pi(\text{par}(u)) = \text{par}_{U(Q)}(\pi(u))$  for every  $u \in T$  with a parent (where  $\text{par}_{U(Q)}$  is walk truncation on  $U(Q)$ );
- (ii)  $\pi$  is child-multiplicity-respecting: for each  $u \in T$  and each  $\sigma' \in S$ , the number of children of  $u$  in  $T$  with  $\text{st}_d$ -image  $\sigma'$  is  $\geq \text{Child}_Q(\text{st}_d(u))(\sigma')$  (with equality up to the  $\{0, 1, 2+\}$  truncation);
- (iii)  $\pi$  is mate-respecting: if  $u, u' \in T$  satisfy  $u \equiv_{\text{EQ}} u'$ , then  $\pi(u), \pi(u') \in U(Q)$  are linked by the  $\equiv_{\text{EQ}, Q}$ -quotient on  $U(Q)$ ;
- (iv)  $\pi$  is surjective on tree-paths: every walk in  $(S, \text{par}_Q)$  that is reachable from  $\sigma_*$  in finitely many steps lies in the image of  $\pi$ .

*Proof.* (i) By Definition 7.3,  $\text{par}_Q(\text{st}_d(u)) = \text{st}_d(\text{par}(u))$ , so  $\pi(\text{par}(u))$  is the walk  $\pi(u)$  truncated by one step.

(ii) The child-multiplicity component of the profile (Definition 7.1, item 4) is extracted from  $T$ :  $\text{Child}_Q(\text{st}_d(u))(\sigma')$  counts the number of children of *any* occurrence of profile  $\text{st}_d(u)$  in  $T$  with  $\text{st}_d$ -image  $\sigma'$ , truncated to  $\{0, 1, 2+\}$ . Because  $u$  itself is one such occurrence,  $u$  has at least  $\text{Child}_Q(\text{st}_d(u))(\sigma')$  such children.

(iii) If  $u \equiv_{\text{EQ}} u'$ , then by (EQ2) of Definition 10.9,  $(\text{st}_d(u), \pi_T(u)) \equiv_{\text{EQ}, Q} (\text{st}_d(u'), \pi_T(u'))$  where  $\pi_T(u)$  is the equality-port tuple of  $u$  (component 2 of the profile). This  $\equiv_{\text{EQ}, Q}$ -link lifts to a quotient on  $U(Q)$  by Definition 14.1.

(iv) Every walk  $(\sigma_*, \sigma_1, \dots, \sigma_m)$  in  $(S, \text{par}_Q)$  corresponds to a chain of profiles each obtained from the previous via  $\text{par}_Q$ . Pick any occurrence  $u \in T$  with  $\text{st}_d(u) = \sigma_m$  and traverse  $\text{par}^k$  for  $k = 0, \dots, m$ ; by (i),  $\pi(u)$  is the walk in question. Such  $u$  exists by definition of  $S$ .  $\square$

*Remark 14.4* ( $T$  is a concrete unfolding witness). By Lemma 14.3,  $T$  is a concrete tree presentation that maps onto  $U(Q)$  via the parent- and child-respecting projection  $\pi$ . We use  $T$  as a witness for realization conditions on  $U(Q)$ : any condition verified pointwise on  $T$  transfers along  $\pi$  to  $U(Q)$ .



**Lemma 14.5** ( $\text{par}_Q$  descends to mate classes). *For all  $\sigma_1, \sigma_2 \in S$ :*

$$\sigma_1 \in \text{Mate}(\sigma_2) \implies \text{par}_Q(\sigma_1) \in \text{Mate}(\text{par}_Q(\sigma_2))$$

(with  $\perp$  on either side meaning both are  $\perp$ ).

*Proof.* Let  $\sigma_1 \in \text{Mate}(\sigma_2)$ , witnessed by ports  $p_1 \in \Pi_{\sigma_1}, p_2 \in \Pi_{\sigma_2}$  with  $(\sigma_1, p_1) \equiv_{\text{EQ}, Q} (\sigma_2, p_2)$ . Pick occurrences  $u_1, u_2 \in T$  with  $\text{st}_d(u_i) = \sigma_i$  and  $\pi(u_i) = p_i$ ; by Lemma 10.11 the typed-EQ congruence  $\equiv_{\text{EQ}}$  satisfies  $u_1 \equiv_{\text{EQ}} u_2$ . Hence  $\text{orig}(u_1)$  and  $\text{orig}(u_2)$  are the same semantic element of  $\mathcal{I}_w$  (strong EQ semantics), and the parent occurrences  $\text{par}(u_1)$  and  $\text{par}(u_2)$  in  $T$  either both project to  $\perp$  (if neither has a generated cover parent) or both have orig image equal to the unique semantic parent in  $\mathcal{I}_w$ . In the latter case  $\text{par}(u_1) \equiv_{\text{EQ}} \text{par}(u_2)$  by (EQ1)–(EQ2), so  $\text{par}_Q(\sigma_1) = \text{st}_d(\text{par}(u_1))$  and  $\text{par}_Q(\sigma_2) = \text{st}_d(\text{par}(u_2))$  lie in the same mate class via the port assignment of Lemma 10.11.  $\square$

**Lemma 14.6** (Simultaneous realization on  $U(Q)$ ). *There is a (possibly infinite) tree presentation  $U^*(Q)$ , concretely realized by the split tree  $T$ , projecting to  $Q$  via  $\text{st}_d$ , such that:*

- (SR1) **(SR1) PP realization.** *For every  $u \in U^*(Q)$  and every  $\exists \text{PP}.D \in \lambda(u)$ , the parent walk  $u, \text{par}(u), \text{par}^2(u), \dots$  visits some occurrence  $u'$  with  $D \in \lambda(u')$ .*
- (SR2) **(SR2) PPI realization.** *For every  $u \in U^*(Q)$  and every  $\exists \text{PPI}.D \in \lambda(u)$ , some descendant occurrence  $u'$  in the child subtree of  $u$  has  $D \in \lambda(u')$ .*
- (SR3) **(SR3) Universal propagation.** *For every  $u$ , every  $\forall R.D \in \lambda(u)$  with  $R \in \{\text{PP}, \text{PPI}\}$ , and every  $u'$  in the parent walk (if  $R = \text{PP}$ ) or child subtree (if  $R = \text{PPI}$ ) reachable from  $u$  through occurrences whose mate class lies in the  $\text{par}_Q$ -reachable mate-class closure of  $\text{st}_d(u)$ :  $D \in \lambda(u')$ .*

*All three conditions hold for the same  $U^*(Q)$  at the same time (“simultaneously”), uniformly over all occurrences  $u$ .*

*Proof.* We take  $U^*(Q) := T$  and verify (SR1)–(SR3) pointwise on  $T$ . Lemma 14.3 (the projection  $\pi : T \rightarrow U(Q)$  is faithful) then transfers the realization conditions from  $T$  to  $U(Q)$  along  $\pi$ .

**(SR1) PP realization.** Let  $u \in T$  with  $\exists \text{PP}.D \in \lambda(u)$ . Since  $\lambda(u) = \text{tp}_{\mathcal{I}_w}(\text{orig}(u))$  and  $\mathcal{I}_w \models C_0$  at every  $\text{orig}(u)$ , there exists  $b \in \mathcal{I}_w$  with  $\rho(\text{orig}(u), b) = \text{PP}$  and  $D \in \text{tp}_{\mathcal{I}_w}(b)$ . Witness generation (Lemma 4.3) forces  $b$  to lie in  $\mathcal{I}_w$ , so  $b$  has at least one occurrence  $u^\dagger \in T$  with  $\text{orig}(u^\dagger) = b$  (Construction 6.1 step 2). Construction 6.1 also installs the cover edge  $(u, u^\dagger)$ : there is a finite chain of generated PP-cover edges

$$u = u_0 \xrightarrow{\text{PP}} u_1 \xrightarrow{\text{PP}} \dots \xrightarrow{\text{PP}} u_n = u^\dagger,$$

each step at most one PP-edge in the witness-generated submodel. The chain has finite length because the cover graph is generated by finitely many witness selections starting at  $a_0$  (Lemma 4.3). Thus the parent walk of  $u$  in  $T$  visits  $u^\dagger$ , and  $D \in \lambda(u^\dagger) = \text{tp}_{\mathcal{I}_w}(b)$ . Projecting along  $\pi$  yields the corresponding  $\text{par}_Q$ -walk in  $U(Q)$ .

**(SR2) PPI realization.** Let  $u \in T$  with  $\exists \text{PPI}.D \in \lambda(u)$ . By the same witness-generation argument, there is  $b' \in \mathcal{I}_w$  with  $\rho(\text{orig}(u), b') = \text{PPI}$  and  $D \in \text{tp}_{\mathcal{I}_w}(b')$ . Construction 6.1 step 3

places an occurrence  $u'' \in T$  as a child of  $u$  with  $\text{orig}(u'') = b'$ . The child multiplicity component  $\text{Child}_Q(\text{st}_d(u))(\text{st}_d(u''))$  is therefore at least 1 (it is the truncated count of such children in  $T$ ). By Lemma 14.3 (ii), the corresponding child slot in  $U(Q)$  also has multiplicity  $\geq 1$ , so the projection  $\pi(u'')$  is a child of  $\pi(u)$  in  $U(Q)$  carrying  $D$  in its profile type.

**(SR3) Universal propagation.** Let  $u \in T$  with  $\forall R.D \in \lambda(u)$ ,  $R \in \{\text{PP}, \text{PPI}\}$ . By (EQ1) of Definition 10.9, every occurrence  $u' \equiv_{\text{EQ}} u$  has  $\forall R.D \in \lambda(u')$ , so the set of profiles in  $\text{Mate}(\text{st}_d(u))$  all carry  $\forall R.D$ . Along any chain of PP- or PPI-cover edges in  $T$  starting from  $u$  (resp. any descendant occurrence of  $u$  via child multiplicity), each edge in  $\mathcal{I}_w$  is an instance of  $R$ , and Lemma 3.2 ensures  $D$  is propagated forward. By Lemma 13.5 (the profile-level statement), the target profile carries  $D$  along the entire  $\text{par}_Q$ - or  $\text{par}_Q^{-1}$ -reachability closure of  $\text{Mate}(\text{st}_d(u))$ . Pulling back along  $\pi$ : every  $u' \in T$  in the parent walk of  $u$  (or descendant child subtree) has  $\lambda(u') = \lambda_Q(\text{st}_d(u')) \ni D$ .

**Simultaneity.** The argument above proceeds occurrence-by-occurrence on a single  $T$ .  $T$  is built once from  $\mathcal{I}_w$  (Construction 6.1); the witness selector  $w$  supplies one witness per existential per occurrence (by Definition 4.1);  $\mathcal{I}_w$  is a single strong-EQ model; hence *all* witnesses coexist in  $T$ . Therefore conditions (SR1)–(SR3) hold for every  $u \in T$  *simultaneously*, witnessed by the same  $T$ .

**Coherence of the rho-labelling on  $U(Q)$ .** The relation labelling  $\rho_T(u, u') = \rho(\text{orig}(u), \text{orig}(u'))$  on pairs  $u, u' \in T$  inherits all RCC5 frame axioms from  $\mathcal{I}_w$  (Section 3). In particular, for any triangle  $(u_1, u_2, u_3)$  of occurrences in  $T$ ,  $\rho_T(u_1, u_3) \in \text{comp}(\rho_T(u_1, u_2), \rho_T(u_2, u_3))$ . This is 3-consistency on the realized triangles of  $T$ . By the path-consistency theorem of Renz–Nebel [1], 3-consistency of an atomic RCC5 network is equivalent to global consistency. Projecting  $\rho_T$  along  $\pi$  to  $U(Q)$  preserves 3-consistency (since the projection is parent- and child-respecting); hence the relation labelling on  $U(Q)$  admits a globally consistent atomic RCC5 refinement. Combined with the existential and universal realizations above, this gives a coherent simultaneous realization on  $U(Q)$ .  $\square$

*Remark 14.7* (O6 discharged). Lemma 14.6 discharges GPT-5.5’s obligation O6. The proof avoids automata-theoretic machinery: it uses only finite-graph reachability (Lemma 13.2), the typed-EQ congruence (Lemma 10.11), and the patchwork property [1]. The PP direction proceeds via the ultimately periodic parent walk in  $(S, \text{par}_Q)$  (where the self-loop case is the period-1 ultimately periodic walk); the PPI direction proceeds via the child-multiplicity component of the profile. Mate-class quantification (introduced in v2.1 for O2) is essential: without it, distinct occurrences carrying distinct ports of the same closure type could have disjoint sets of realized witnesses, breaking simultaneity.

*Remark 14.8* (Caveat:  $T$  is not unique).  $T$  depends on the choice of witness selector  $w$  (Definition 4.1). Different selectors yield different concrete unfoldings, all projecting to the same  $Q$  and all satisfying (SR1)–(SR3). Lemma 14.6 asserts existence, not uniqueness; the existence of *some*  $U^*(Q)$  is what the validity argument requires. The abstract canonical unfolding  $U(Q)$  of Definition 14.1 is one specific witness (modulo equality-port quotienting).

## 15 Validity of the extracted quotient

We verify that the quotient extracted in Definition 7.3 is valid in the sense of [5, Definition 1.10] (or equivalently of [6, Definition 8.2]).

**Theorem 15.1** (Validity). *The quotient  $Q$  extracted from  $(\mathcal{I}_w, a_0)$  in Definition 7.3 satisfies all validity conditions:*

- (V1) every existential  $\exists R.D \in \lambda_Q(\sigma)$  has a witness;
- (V2) every universal  $\forall R.D \in \lambda_Q(\sigma)$  is respected;
- (V3) every domain in the quotient has non-empty  $\text{Need}_R$ -filtered domain;
- (V4) the extracted descriptors are support-closed and locally coherent;
- (V5)  $\equiv_{\text{EQ}}$  is a typed-EQ congruence;
- (V6)  $C_0 \in \lambda_Q(\sigma_*)$ .

*Proof.* **(V1) Existential satisfaction.** For  $R \in \{\text{DR}, \text{PO}\}$ :  $\text{Wit}_d(\sigma)$  provides  $(\sigma', R)$  with  $D \in \lambda_Q(\sigma')$ , by Construction 9.1. Relation-safety holds by Lemma 9.2.

For  $R \in \{\text{PP}, \text{PPI}\}$ : by Lemma 13.2 (v2.1), there exist  $\hat{\sigma} \in \text{Mate}(\sigma)$  and  $\sigma' \in S$  with  $\hat{\sigma} \rightsquigarrow_R \sigma'$  and  $D \in \lambda_Q(\sigma')$ . The certificate's discharge predicate quantifies over  $\text{Mate}(\sigma)$  as a single source, so this satisfies (V1). Simultaneous realization of all such eventualities at all occurrences in a single coherent unfolding follows from Lemma 14.6.

**(V2) Universal satisfaction.** For  $R \in \{\text{DR}, \text{PO}\}$ : any sibling-mosaic edge from  $\sigma$  to  $\sigma'$  labelled  $R$  is realized by a model pair, which satisfies the universal by Lemma 3.2.

For  $R \in \{\text{PP}, \text{PPI}\}$ : Lemma 13.5 (v2.1) establishes universal propagation along parent reachability from every  $\hat{\sigma} \in \text{Mate}(\sigma)$ . Hence the universal is preserved under all mate-aware reachability, which is the (V2) condition for the equality-port-extended certificate.

**(V3) Non-empty filtered domains.** For any pair-domain  $\text{Dom}(\sigma_1, \sigma_2)$  extracted from the model:

$$\text{Dom}(\sigma_1, \sigma_2) = \{\rho(\text{orig}(u), \text{orig}(v)) : \text{st}_d(u) = \sigma_1, \text{st}_d(v) = \sigma_2\}.$$

Every  $R \in \text{Dom}(\sigma_1, \sigma_2)$  is realized by a model pair, hence is relation-safe by Lemma 3.2. Therefore the  $\text{Need}_R$ -filtered domain equals the unfiltered domain, and is non-empty whenever  $\text{Dom}(\sigma_1, \sigma_2)$  is non-empty.

**(V4) Coherent descriptors.** By Theorem 11.1, with effective enumerability guaranteed by Theorem 12.4 (locality at arity  $N(C_0) = 5$ ).

**(V5) Typed-EQ congruence.** By Lemma 10.11.

**(V6) Evaluation state.**  $u_*$  has  $\text{orig}(u_*) = a_0$  and  $\lambda(u_*) = \text{tp}_{\mathcal{I}_w}(a_0) \ni C_0$ . Hence  $C_0 \in \lambda_Q(\sigma_*)$ .  $\square$

## 16 Main theorem and decidability

*Proof of Theorem 1.1.* Given  $\mathcal{I}, d_* \models C_0$ :

1. Choose a selected witness policy  $w$  (Definition 4.1).

2. Pass to the witness-generated submodel  $\mathcal{I}_w$  (Lemma 4.3);  $a_0 := d_* \in C_0^{\mathcal{I}_w}$ .
3. Construct the generated split tree  $T$  (Construction 6.1, Lemma 6.3).
4. Define rank- $d$  profiles  $\text{st}_d$  (Definition 7.1).
5. Form the finite quotient  $Q$  (Definition 7.3).
6. Extract sibling descriptors as support-closed mosaics (Construction 8.8, Lemma 8.9).
7. Extract witness menus (Construction 9.1, Lemma 9.2).
8. Verify validity (Theorem 15.1).

The resulting  $Q$  is a valid finite generated split-forest quotient for  $C_0$  with  $C_0 \in \lambda_Q(\sigma_*)$ .  $\square$

*Proof of Corollary 1.2.*  $|\Sigma_d|$  is finite by Lemma 7.2. Validity of a candidate quotient  $Q$  is decidable: each (V1)–(V6) condition is a finite check on finite data; reachability (V1) for PP/PPI is decidable in  $O(|S|^2)$  per eventuality; descriptor coherence (V4) is decidable in time polynomial in  $|\text{Mos}|$ .

If  $C_0$  is satisfiable, Theorem 1.1 produces a valid quotient. If a valid quotient exists, soundness ([5, Theorem 1.20], [6, Theorem 9.1]) produces a model. Thus  $C_0$  is satisfiable iff a valid finite quotient exists, and enumeration terminates.  $\square$

## 17 Discussion

### 17.1 Why this is not Büchi acceptance

The key observation that lets us avoid automata-theoretic machinery: every transitive eventuality  $\exists R.D$  at  $\sigma \in S$  for  $R \in \{\text{PP}, \text{PPI}\}$  is discharged by ordinary reachability over the finite directed graph  $(S, \text{par}_Q)$ .

In an unfolding of  $Q$ , an infinite PP-branch corresponds to a walk in  $(S, \text{par}_Q)$ . Once the walk enters a strongly connected component (SCC), it cycles through that component forever (because  $|S|$  is finite). Eventually-recurrent profiles on the SCC are exactly the profiles reachable from any profile in the SCC.

So: a PP-eventuality  $\exists \text{PP}.D$  at  $\sigma$  is realized in the unfolding iff  $D$  appears in some profile reachable from  $\sigma$ . This is decidable by:

1. BFS from  $\sigma$  in  $(S, \text{par}_Q)$ ;
2. check whether any reachable profile  $\sigma'$  has  $D \in \lambda_Q(\sigma')$ .

GPT-5.5’s stem-and-cycle phrasing (“ $u_0, \dots, u_m, (v_0, \dots, v_{k-1})^\omega$ ”, [6, Definition 6.4]) is one specific way to present this realization condition. But the underlying check is the same finite reachability question, and our formulation in Lemma 13.2 is closer to standard description logic blocking conditions.

## 17.2 Comparison with v1

| Concern               | v1 (April 2026)                                      | v2 (this paper)                     |
|-----------------------|------------------------------------------------------|-------------------------------------|
| Starting model        | arbitrary PP-DAG with assumed $\leq_{\text{PP-min}}$ | witness-generated submodel          |
| Cover edges           | Hasse covers of ambient order                        | generated cover edges in $D_w(a_0)$ |
| Ambient root          | required (added freshly if needed)                   | not required                        |
| Self-loops on parent  | not permitted                                        | permitted (handles $C \uparrow$ )   |
| Sibling descriptor    | raw pair-tables                                      | support-closed mosaics              |
| Triple coherence (C5) | witness conflation across occurrences                | joint-realization within a mosaic   |
| Typed-EQ congruence   | implicit via orig map                                | explicit axiom                      |
| PP/PPI eventualities  | “ancestor in tree” (vague)                           | finite graph reachability           |

## 17.3 Comparison with GPT-5.5’s automata version

GPT-5.5’s coherent generated split-forest proof [6] arrives at the same conclusion via the same structural ideas (witness-generated submodels, generated covers, support-closed mosaics, typed-EQ congruence) but formalizes the eventuality discharge as a Büchi-style cycle-realization condition.

Our v2 differs only in presentation: where GPT-5.5 writes  $u_0 \cdots u_m(v_0 \cdots v_{k-1})^\omega$  and requires every eventuality on the cycle to be met on the cycle, we write “every  $\exists \text{PP}.D$  at  $\sigma$  has a reachable witness profile,” which is finite-graph reachability. These are equivalent: in the finite quotient, “met somewhere on the  $\omega$ -trace” equals “reachable in  $\text{par}_Q$ .”

The benefit of the reachability formulation: it stays inside the standard description-logic toolbox (similar to fixpoint blocking conditions in  $\mu$ -calculus tableaux), and avoids invoking  $\omega$ -language theory.

## 17.4 Honest assessment

We flag the following points for scrutiny:

**Witness policy is non-canonical.** Different selected witness policies produce different generated submodels and different split trees. The extraction works for *any* fixed policy. We do not claim a canonical choice.

**Profile bound is not elementary.**  $|\Sigma_d|$  is bounded by a tower of exponentials in  $|\text{cl}(C_0)|$ . Decidability follows, but tight complexity remains open. The PO-coherent fragment achieves a 2-EXPTIME quotient bound [10].

**Mosaic vertex bound.** Construction 8.8 restricts to mosaics on  $\leq |\text{cl}(C_0)| + 4$  vertices. This is sufficient for support-closure (SC1)–(SC4), but a fully rigorous treatment would derive the bound from a Ramsey-type argument on closure size. GPT-5.5 sketches such an argument in [6, Lemma 8.4].

**Reachability vs strongly-connected components.** Lemma 13.2 establishes existence of a reachable witness profile for each eventuality. The full correctness argument requires that *simultaneous* discharge of all eventualities on a branch is realized by a single coherent infinite walk; this is GPT-5.5’s obligation O6, discharged in v2.1 by Lemma 14.6. The argument uses finiteness of  $S$ , mate-class quantification, and the RCC5 patchwork property [1]: the SCC of  $\sigma$  in  $(S, \text{par}_Q)$  contains all profiles eventually visited from  $\sigma$ , and any ordering of the eventualities can be interleaved on the SCC walk.

None of these points appears to harbour a genuine gap. The extraction rests on three clean principles:

1. witness-generated submodels are sparse (Lemma 4.3);
2. model edges are relation-safe (Lemma 3.2);
3. finite-graph reachability discharges transitive eventualities (Lemma 13.2).

## 18 Conclusion

We have written out the completeness extraction for  $\mathcal{ALCI}_{\text{RCC5}}$  in a repaired form that fixes the two structural defects of the v1 draft: the unsound Hasse construction and the witness-conflated triple-coherence proof. The repair uses witness-generated submodels, generated cover edges, support-closed mosaics, an explicit typed-EQ congruence, and finite-graph reachability discharge.

Crucially, the repair stays inside the original split-forest framework: vertical PP/PPI branches, sibling-interface descriptors, and finite quotient enumeration are all preserved. The cycle-realization condition that motivates GPT-5.5’s automata-flavoured proof reduces here to ordinary reachability over the finite quotient graph, sidestepping any need for Büchi acceptance or  $\omega$ -language theory.

Combined with the soundness chain of [5, 6], this yields:

*Concept satisfiability in  $\mathcal{ALCI}_{\text{RCC5}}$  under strong EQ semantics is decidable.*

The exact complexity is open. The quotient search is decidable but non-elementary; an EXPTIME bound (matching the known  $\mathcal{ALCI}$  lower bound) would require a tighter analysis of the rank- $d$  profile space.

**Computational evidence.** An independent implementation [11] cross-validates the cover-tree tableau (the operational counterpart of the split-forest quotient search) against a quasimodel-based reasoner on 911 test concepts with zero mismatches. A decomposition test confirms that 775/775 satisfiable concepts (100%) have finite cover-tree models across 12.2 M enumerated models, with zero genuine counterexamples [12, 13].

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